Point-spread Functions in Tomography using Filtered Back-projection Reconstruction

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Preface

The work presented in this thesis was carried out from March 1998 to February 2000. During that time, the author also taught in high school, served civil compulsory service, became a husband and stayed a father. I would like to express my deep thanks for the acceptance given to me by my family and friends, this thesis would never have been completed without that understanding support. Also, I would like to thank my supervisor associate professor Stig Steenstrup, who has been an invaluable source of information and support. Also, I would like to thank my fellow students Anders Mortensen, Kim Austin Smith, Peter Harald Lindenov Christiansen, Erik Mejdal Lauridsen, Claus Tilsted Mogensen and Lars Charmer Jørgensen. University life would have been all dull and gray without your company. Finally, I thank Kenneth Geisshirt, Peter Harald Lindenov Christiansen, Anders Mortensen and Claus Tilsted Mogensen for the time they spent proof reading this thesis.

"Life is what happens, while you are busy making other plans"
- John Lennon

This thesis is dedicated to my wife and children - born and unborn.

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Chapter 1

Introduction

Ever since the discovery of the penetrating abilities of X-ray radiation [5] by Wilhelm Röntgen, many scientists have put a lot of work into the field of medical imaging. The fantastic ability to investigate the interior of a biological sample or even a medical patient without the use of surgery has been perfected, using new knowledge from physics as well as many great technical achievements. With the invention and perfection of computers, the possibility to reconstruct a function (e.g. the attenuation coefficient of the tissue in a slice of a human flesh or perhaps the regional cerebral blood flow of an activated brain) from its projections was suddenly possible. A whole new field of medical imaging techniques emerged. Techniques such as CAT or CT, SPECT and PET were developed, all on the basis of the mathematical work done by J. Radon in 1917 [1]. The physical and technical details of these scanner types are discussed in chapter 3.

As these techniques rapidly evolved, the need to quantify and describe the differences and mutualities of the different techniques grew. It is perfectly clear that, as in all fields of science, the limitations of the used model or theory is very important to understand. Hence, a collection of scientific papers [16] – [19] on this topic were written, but all of these were specific to one of the different imaging techniques based on Radon's mathematical work. One of the main targets of this thesis is to quantify the relative error or derivation from a perfectly reconstructed point source, (in other words the point-spread function) as a function of different parameters, *without* reference to a specific imaging technique. An unpublished, theoretical paper [15] by Stig Steenstrup takes the same approach, and claims to have found a common, best possible point-spread function for tomographic imaging. The paper shows how this function depends on a set of reconstruction parameters.

My work has been to investigate the validity of the theory presented in [15], using the 3D-reconstruction method of *filtered back-projection* (FBP),

as described in section 2.3. Point-spread functions are investigated through simulated data, which are reconstructed by use of an FBP computer-program developed by the author, for specific use in this thesis. The presented data shows how the point-spread function of FBP depends on the same set of parametes, investigated in [15]. Specifically, dependence of the point-spread function width on several parameters is investigated.

In the thesis, a thorough explanation of the general mathematics and technology of modern tomography scanners is given, and the reconstruction method of FBP is explained in detail. Two examples of the use of the underlying mathematics are given, to illustrate the complexity of using Radon's theory for analytical reconstructions. Three medical imaging techniques (CT, SPECT and PET) are covered in chapter 3. The presented theory, the analytical reconstruction examples and the description of scanner technology together form a good overview of the field of medical scanners, suitable for an unfamiliar reader.

The thesis is composed of four parts. The first part describes the general theory of reconstruction from a mathematical and technical point of view. The second part covers reconstruction of point sources, including the theory presented in [15] as well as the details of the simulation study. The third part of the thesis analyses the results found, and the fourth part is a conclusion and perspectivation, based on the results found.

Chapter 2

Mathematics of the Radon transform

In the year 1917, the Austrian mathematician Johann Radon published a paper [1], which explained how a real function of two real variables could be reconstructed from its line integrals. The paper of Radon is the founding mathematics of many reconstruction methods. In this chapter we shall investigate the work of Radon in some detail.

2.1 Derivation of the Radon inversion formula

In this section we will derive the reconstruction formula as proposed by Johann Radon.

2.1.1 Line integrals - projections

Let $f(x, y)$ be a well behaved[1] complex function of two real variables (x, y). We are then able to define a set of line integrals or projections of f as

$$g(s, \theta) = \int_{l_{s,\theta}} f(x, y) dl, \tag{2.1}$$

where

$$l_{s,\theta} = \begin{pmatrix} x(u) \\ y(u) \end{pmatrix} = \begin{pmatrix} s\cos\theta - u\sin\theta \\ s\sin\theta + u\cos\theta \end{pmatrix} \tag{2.2}$$

[1] It is assumed that f has all needed properties in each given case.

is the set of points in $\mathbb{R}^2$ that are members of the line parallel to the direction rotated by θ from the x-axis and situated at a distance s from the origin. Insertion of (2.2) into (2.1) yields

$$[\mathcal{R}f](s,\theta) = g(s,\theta) = \int_{-\infty}^{\infty} f(s\cos\theta - u\sin\theta, s\sin\theta + u\cos\theta)du. \tag{2.3}$$

We name $[\mathcal{R}f]$ or g the *Radon-transform* of f. Knowing all possible g's, we are able to reconstruct f, using the Radon inversion formula, in the version presented in Radon's original paper [1]

$$f(r,\phi) = -\frac{1}{2\pi^2}\int_{-\infty}^{\infty}\int_{0}^{\pi} \frac{1}{s - r\cos(\theta-\phi)}\frac{\partial g(s,\theta)}{\partial s}d\theta ds. \tag{2.4}$$

Later, an equivalent formula was established, using Fourier theory

$$f(x,y) = [\mathcal{B}\{\mathcal{F}_1^{-1}(|S| \times \mathcal{F}_1 g)\}](x,y). \tag{2.5}$$

A proof of this formulation is carried out in a later section.

2.1.2 Needed transformations

To establish the Radon reconstruction formulas, we must define a number of Fourier transform identities.
For functions $f : \mathbb{R} \to \mathbb{C}$, the Fourier transform and inverse are given by

$$[\mathcal{F}f](X) = \int_{\mathbb{R}} f(x)e^{-ixX}dx \tag{2.6}$$

$$[\mathcal{F}^{-1}\tilde{f}](x) = \frac{1}{2\pi}\int_{\mathbb{R}} \tilde{f}(X)e^{ixX}dX \tag{2.7}$$

For functions $f : \mathbb{R}^2 \to \mathbb{C}$, the Fourier transform and inverse are given by

$$[\mathcal{F}_2 f](X,Y) = \int_{\mathbb{R}^2} f(x,y)e^{-i(xX+yY)}dxdy \tag{2.8}$$

$$[\mathcal{F}_2^{-1}\tilde{f}](x,y) = \frac{1}{4\pi^2}\int_{\mathbb{R}^2} \tilde{f}(X,Y)e^{i(xX+yY)}dXdY \tag{2.9}$$

For functions $g : \mathbb{R} \times [0, 2\pi] \to \mathbb{C}$, the (one-dimensional) Fourier transform and inverse are given by

$$[\mathcal{F}_1 g](S,\theta) = \int_{\mathbb{R}} g(s,\theta)e^{-isS}ds \tag{2.10}$$

$$[\mathcal{F}_1^{-1}\tilde{g}](s,\theta) = \frac{1}{2\pi}\int_{\mathbb{R}} \tilde{g}(S,\theta)e^{isS}dS \tag{2.11}$$

2.1.3 The Projection-formula

In this section we use the previous definitions to prove the Radon projection-formula, enabling us to reconstruct f from the set of g's

$$[\mathcal{F}_1\mathcal{R}f](S,\theta) = [\mathcal{F}_2 f](S\cos\theta, S\sin\theta). \tag{2.12}$$

Let us begin by inserting bits and pieces on the right side of (2.12)

$$[\mathcal{F}_1\mathcal{R}f](S,\theta) = \int\int f(s\cos\theta - t\sin\theta, s\sin\theta + t\cos\theta)e^{-isS}dtds.$$

Changing coordinates according to (2.2) we get (the Jacobian is 1 during rotation)

$$\begin{aligned}
[\mathcal{F}_1\mathcal{R}f](S,\theta) &= \int\int f(x,y)e^{-i(x\cos\theta + y\sin\theta)S}dxdy \\
&= \int\int f(x,y)e^{-i(x(S\cos\theta)+y(S\sin\theta))}dxdy \\
&= [\mathcal{F}_2 f](S\cos\theta, S\sin\theta)
\end{aligned}$$

Hence we can extract f from the set of projections by the transformation relation

$$[\mathcal{F}_1\mathcal{R}f](S,\theta) = [\mathcal{F}_2 f](S\cos\theta, S\sin\theta).$$

Applying $\mathcal{F}_2^{-1}$

$$f(x,y) = [\mathcal{F}_2^{-1}\mathcal{F}_1\mathcal{R}f](S,\theta) = [\mathcal{F}_2^{-1}\mathcal{F}_1 g](s,\theta).$$

2.1.4 The Radon inversion formula - Fourier formulation

In this section we extract an explicit reconstruction formula from the above projection formula. By definition we have that

$$f(x,y) = \mathcal{F}_2^{-1}\mathcal{F}_2 f(x,y)$$

$$= \frac{1}{4\pi^2}\int_{\mathbb{R}^2}[\mathcal{F}_2 f](X,Y)e^{i(xX+yY)}dXdY.$$

Changing coordinates such that $X = S\cos\theta$ and $Y = S\sin\theta$, yields

$$f(x,y) = \frac{1}{4\pi^2}\int_0^{2\pi}\int_0^{\infty}[\mathcal{F}_2 f](S\cos\theta, S\sin\theta)e^{iS(x\cos\theta + y\sin\theta)}SdSd\theta$$

$$= \frac{1}{4\pi^2} \int_0^{\pi} \int_{-\infty}^{\infty} [\mathcal{F}_2 f](S\cos\theta, S\sin\theta) e^{iS(x\cos\theta + y\sin\theta)} |S| dS d\theta,$$

since the Jacobian becomes $J = S$. Using the projection formula (2.12), we obtain

$$f(x,y) = \frac{1}{4\pi^2} \int_0^{\pi} \int_{-\infty}^{\infty} [\mathcal{F}_1 \mathcal{R} f](S\cos\theta, S\sin\theta) e^{iS(x\cos\theta + y\sin\theta)} |S| dS d\theta$$

$$= \frac{1}{2\pi} \int_0^{\pi} [\mathcal{F}_1^{-1}(|S| \times \mathcal{F}_1 \mathcal{R} f)](x\cos\theta + y\sin\theta, \theta) d\theta.$$

$$= \mathcal{B}(\mathcal{F}_1^{-1}(|S| \times \mathcal{F}_1 \mathcal{R} f))$$

$$= \mathcal{B}(\mathcal{F}_1^{-1}(|S| \times \mathcal{F}_1 g))$$

The operator $\mathcal{B}$ will be referred to as the *back-projection operator*, and is defined as

$$[\mathcal{B}\tilde{g}](x,y) = \frac{1}{2\pi} \int_0^{\pi} \tilde{g}(x\cos\theta, y\sin\theta) d\theta. \tag{2.13}$$

2.2 Examples

To illustrate that using the result of Radon in practice is not a trivial task, two calculations for simple functions are completed below. The examples are solved using the two equivalent formulations of the Radon reconstruction formula.

2.2.1 Constant

Let us imagine that a function $f(x,y)$ is given by

$$f(x,y) = \begin{cases} 1, & \text{if } x^2 + y^2 < 1 \\ 0, & \text{elsewhere} \end{cases}$$

We will start out by determining the shape of the projections of f:

$$g(s,\theta) = \int_{-\infty}^{\infty} f(s\cos\theta - u\sin\theta, s\sin\theta + u\cos\theta) du$$

From the definition of f, contributions to the integral in g only occur when $u^2 + s^2 < 1$, meaning that

$$g(s,\theta) = \int_{-\sqrt{1-s^2}}^{\sqrt{1-s^2}} f(s\cos\theta - u\sin\theta, s\sin\theta + u\cos\theta) du,$$

which simply is

$$g(s,\theta) = \int_{-\sqrt{1-s^2}}^{\sqrt{1-s^2}} du = 2\sqrt{1-s^2}.$$

The original form of the Radon inversion formula is

$$f(r,\phi) = -\frac{1}{2\pi^2}\int_{-\infty}^{\infty}\int_{0}^{\pi} \frac{1}{s - r\cos(\theta-\phi)}\frac{\partial g(s,\theta)}{\partial s} d\theta ds.$$

The derivative of $g(s,\theta)$ with respect to s is

$$\frac{\partial g(s,\theta)}{\partial s} = -\frac{2s}{\sqrt{1-s^2}},$$

which leads to

$$f(r,\phi) = \frac{1}{\pi^2}\int_{-\infty}^{\infty}\int_{0}^{\pi} \frac{1}{s - r\cos(\theta-\phi)}\frac{s}{\sqrt{1-s^2}} d\theta ds.$$

Changing θ integration limits to $[0, 2\pi]$ changes a sign, which is resolved by changing s integration limits to $[0,\infty]$:

$$f(r,\phi) = \frac{1}{\pi^2}\int_{0}^{\infty}\int_{0}^{2\pi} \frac{1}{s - r\cos(\theta-\phi)}\frac{s}{\sqrt{1-s^2}} d\theta ds.$$

From the definition of g, the integrand is only non-zero when $s < 1$:

$$f(r,\phi) = \frac{1}{\pi^2}\int_{0}^{1} \frac{s}{\sqrt{1-s^2}}\left\{\int_{0}^{2\pi} \frac{1}{s - r\cos(\theta-\phi)} d\theta\right\} ds \qquad (2.14)$$

The θ-integration is

$$I = \int_{0}^{2\pi} \frac{d\theta}{s - r\cos(\theta-\phi)} = \frac{1}{i}\oint \frac{dz}{z(s - \frac{1}{2}r(z+z^{-1}))} = \frac{2}{-ri}\oint \frac{dz}{z^2 - 2az + 1},$$

with $z = e^{i\theta}$ and $a = s/r$. Let us now consider two cases:

a) When $a > 1$ $(s > r)$ the integral becomes

$$I = \frac{2}{-ri}\oint \frac{dz}{(z-z_1)(z-z_2)},$$

with $z_1 = a + \sqrt{a^2+1}$ and $z_2 = a - \sqrt{a^2-1}$. Using the Cauchy residual theorem, we obtain

$$I = 2\pi i \frac{2}{-ri}\frac{1}{(z2 - z_1)} = 2\pi \frac{1}{-r\sqrt{a^2-1}} = \frac{2\pi}{\sqrt{s^2-r^2}}.$$

b) When $a < 1$ $(s < r)$, I is

$$I = \frac{2}{-ri} \oint \frac{dz}{(z - z_1)(z - z_2)}, \tag{2.15}$$

with $z_1 = a + i\sqrt{1 - a^2}$ and $z_2 = a - i\sqrt{1 - a^2}$. The situation is illustrated in figure 2.1. If one examined (2.15) in great detail using

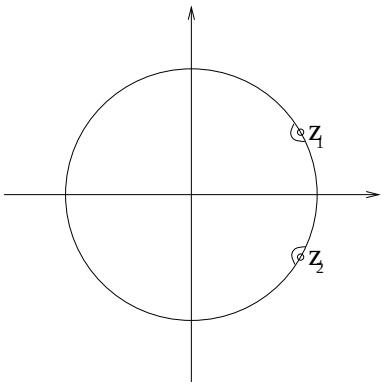


Figure 2.1: Poles from (2.15).

complex analysis, one would find that the contributions from each of the poles shown in figure 2.1 are of equal size, but with different signs, so that the integral in (2.15) becomes 0.

Summarizing, I is non-zero only in case **a** where $s > r$, so (2.14) reduces to

$$f(r, \phi) = \frac{1}{\pi^2} \int_r^1 \frac{s}{\sqrt{1 - s^2}} \frac{2\pi}{\sqrt{r^2 - s^2}} ds = \frac{2}{\pi} \int_r^1 \frac{s}{\sqrt{1 - s^2}\sqrt{r^2 - s^2}} ds.$$

Substituting $z = s^2$, we obtain

$$f(r, \phi) = \frac{1}{\pi} \int_{r^2}^1 \frac{1}{\sqrt{1 - z}\sqrt{r^2 - z}} dz.$$

Next, substituting $t = \frac{z - r^2}{1 - r^2}$, we obtain

$$f(r, \phi) = \frac{1}{\pi} \int_0^1 t^{-1/2}(1 - t)^{-1/2} dt = \frac{1}{\pi} B(\frac{1}{2}, \frac{1}{2}).$$

Inserting the relationship of B with the Γ-function we find that

$$f(r, \phi) = \frac{1}{\pi} \frac{\Gamma(\frac{1}{2})\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2} + \frac{1}{2})} = \frac{1}{\pi} \frac{\sqrt{\pi}\sqrt{\pi}}{1} = 1,$$

when $r < 1$, and from above 0 elsewhere, in perfect agreement with the original definition of f.

2.2.2 Gaussian bell

Let $f(x, y)$ be defined by

$$f(x,y) = \exp\left\{-\frac{(x-x_0)^2}{2\sigma^2} - \frac{(y-y_0)^2}{2\sigma^2}\right\} \tag{2.16}$$

The Radon transform of f is

$$g(s,\theta) = \int_{-\infty}^{\infty} f(x,y)du = \int_{-\infty}^{\infty} \exp\left\{-\frac{(x-x_0)^2}{2\sigma^2} - \frac{(y-y_0)^2}{2\sigma^2}\right\} du \tag{2.17}$$

We now take a closer look at the positive version of the argument of f:

$$\frac{1}{2\sigma^2}\left((x-x_0)^2 + (y-y_0)^2\right)$$

Let us ignore the $\frac{1}{2\sigma^2}$. Only x and y depend on u, inserting this dependency we get:

$$x^2 + x_0^2 - 2xx_0 + y^2 + y_0^2 - 2yy_0 =$$

$$s^2\cos^2\theta + u^2\sin^2\theta + x_0^2 - 2x_0x + s^2\sin^2\theta + u^2\cos^2\theta + y_0^2 - 2y_0y =$$

$$s^2 + u^2 + x_0^2 + y_0^2 - 2x_0(s\cos\theta - u\sin\theta) - 2y_0(s\sin\theta + u\cos\theta) =$$

Using addition formulae for trigonometric functions we obtain

$$u^2 + 2r_0u\sin(\theta-\theta_0) - 2r_0s\cos(\theta-\theta_0) + s^2 + r_0^2$$

This means that the positive argument of the exponential is:

$$\frac{1}{2\sigma^2}u^2 + 2\frac{1}{2\sigma^2}r_0u\sin(\theta-\theta_0) - 2\frac{1}{2\sigma^2}r_0s\cos(\theta-\theta_0) + \frac{1}{2\sigma^2}s^2 + \frac{1}{2\sigma^2}r_0^2 =$$

$$au^2 + bu + c,$$

with a, b and c given by:

$$a = \frac{1}{2\sigma^2}, \quad b = \frac{1}{2\sigma^2}r_0\sin(\theta-\theta_0),$$

$$c = \frac{1}{2\sigma^2}\left\{2r_0\cos(\theta-\theta_0) + s^2 + r_0^2\right\}$$

and g become

$$g(s,\theta) = \int_{-\infty}^{\infty} \exp\left\{-(au^2 + bu + c)\right\} du. \tag{2.18}$$

According to [22] (15.75), p.98, the solution to the integration is:

$$g(s,\theta) = \sqrt{\frac{\pi}{a}} \exp\left\{\frac{b^2 - 4ac}{4a}\right\}$$

Inserting the correct a, b and c, the argument of the exponential in g becomes

$$\frac{1}{2\sigma^2}\left(r_0^2 \sin^2(\theta - \theta_0) - s^2 - r_0^2 + 2r_0 \cos(\theta - \theta_0)\right),$$

which can be simplified to

$$-\frac{(r_0 \cos(\theta - \theta_0) - s)^2}{2\sigma^2}.$$

With this result the g's have the form

$$g(s,\theta) = \sqrt{2\pi}\sigma \exp\left\{-\frac{(r_0 \cos(\theta - \theta_0) - s)^2}{2\sigma^2}\right\}. \tag{2.19}$$

Note that because of the circular symmetry, g only depends on the *magnitude* r_0 of the displacement from the origin, and only on the angular *difference* $\theta - \theta_0$.
Now that the shape of the projections is known to us, let us see if (2.12) is fulfilled. First, let us evaluate the left hand side of that equation:

$$G(S,\theta) = \int_{-\infty}^{\infty} g(s,\theta) e^{isS} ds$$

Inserting g we get:

$$G(S,\theta) = \sqrt{2\pi}\sigma \int_{-\infty}^{\infty} \exp\left\{-\frac{(r_0 \cos(\theta - \theta_0) - s)^2}{2\sigma^2}\right\} e^{isS} ds$$

Expanding the exponential arguments we find that

$$-\frac{1}{2\sigma^2}\left(s^2 + r_0^2 \cos^2(\theta - \theta_0) - 2sr_0 \cos(\theta - \theta_0)\right) + isS =$$

$$-\frac{1}{2\sigma^2}\left\{\left(s^2 + r_0^2 - 2s\left(r_0 \cos(\theta - \theta_0) + \sigma^2 iS\right)\right) + r_0^2 \cos^2(\theta - \theta_0)^2\right\} =$$

$$-\frac{1}{2\sigma^2}\left\{\left(s - (r_0 \cos(\theta - \theta_0) + \sigma^2 iS)\right)^2 - (r_0 \cos(\theta - \theta_0) + \sigma^2 iS)^2 + r_0^2 \cos^2(\theta - \theta_0)\right\} =$$

$$-\frac{(s - (r_0 \cos(\theta - \theta_0) + \sigma^2 iS))^2}{2\sigma^2} - \frac{1}{2}\sigma^2 S^2 + iSr_0 \cos(\theta - \theta_0)$$

This means that G becomes

$$G(S,\theta) = \sqrt{2\pi}\sigma e^{-\frac{1}{2}\sigma^2 S^2} e^{iSr_0\cos(\theta-\theta_0)} \int_{-\infty}^{\infty} \exp\left\{-\frac{(s-(r_0\cos(\theta-\theta_0)+\sigma^2 iS))^2}{2\sigma^2}\right\} ds.$$

With $t = s - (r_0\cos(\theta-\theta_0)+\sigma^2 iS)$, G simplifies to

$$G(S,\theta) = \sqrt{2\pi}\sigma e^{-\frac{1}{2}\sigma^2 S^2} e^{iSr_0\cos(\theta-\theta_0)} \int_{-\infty}^{\infty} \exp\{-\frac{t^2}{2\sigma^2}\} dt,$$

which evaluates to

$$G(S,\theta) = 2\pi\sigma^2 e^{-\frac{1}{2}\sigma^2 S^2} e^{iSr_0\cos(\theta-\theta_0)}.$$

According to the Radon projection formula,

$$G(S,\theta) = F(X,Y) = [\mathcal{F}_2 f](X,Y)$$

Hence, f should be reconstructed by evaluating

$$f(x,y) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(X,Y) e^{-i(xX+yY)} dX\, dY.$$

Using that $X = S\cos\theta$, $Y = S\sin\theta$, $x_0 = r_0\cos\theta_0$ and $y_0 = r_0\sin\theta_0$, the imaginary exponential in G becomes

$$Sr_0\cos(\theta-\theta_0) = Sr_0(\cos\theta\cos\theta_0 + \sin\theta\sin\theta_0) = x_0X + y_0Y.$$

Also inserting that $S^2 = X^2 + Y^2$, $F(X,Y)$ is

$$F(X,Y) = 2\pi\sigma^2 e^{\frac{-\sigma^2}{2}(X^2+Y^2)} e^{i(x_0X+y_0Y)}$$

This means that finding f becomes easier, as the integration splits up:

$$f(x,y) = \frac{1}{2\pi}\sigma^2 \int_{-\infty}^{\infty} e^{\frac{-\sigma^2}{2}X^2} e^{-i(x-x_0)X} dX \int_{-\infty}^{\infty} e^{\frac{-\sigma^2}{2}Y^2} e^{-i(y-y_0)Y} dY.$$

Completing the square of the exponentials in the X-integral, the arguments become

$$-\frac{\sigma^2}{2}(X^2 + i\frac{2}{\sigma^2}(x-x_0)X) =$$

$$-\frac{\sigma^2}{2}((X + i\frac{1}{\sigma^2}(x-x_0))^2 + \frac{1}{\sigma^4}(x-x_0)^2),$$

which makes the x-integration simple:

$$e^{-\frac{(x-x_0)^2}{2\sigma^2}} \int_{-\infty}^{\infty} e^{\frac{-\sigma^2}{2}\tilde{X}^2} d\tilde{X} = e^{-\frac{(x-x_0)^2}{2\sigma^2}} \frac{\sqrt{2\pi}}{\sigma},$$

with $\tilde{X} = X + i\frac{1}{\sigma^2}(x - x_0)$. A similar result is obtained for the Y-integration, and finally f becomes

$$f(x,y) = \exp\left\{-\frac{(x-x_0)^2}{2\sigma^2} - \frac{(y-y_0)^2}{2\sigma^2}\right\},$$

which exactly matches our starting point.

2.3 Practical implementation - filtered back-projection

The reconstruction formulas (2.4) and (2.5) above require modification for practical use, as for example *all* projections must be known prior to application. This is not possible in practice, as only a finite number of detectors and directions of view are used, hence only a low number of line-integrals $g(s,\theta)$ are known. Only a finite set of detector positions $-1 \leq s_j \leq 1$ with $1 \leq j \leq J$ are known for each value $0 \leq \theta_n \leq 2\pi$ with $1 \leq n \leq N$. The situation is illustrated in figure 2.2. To cope with this problem, a discrete

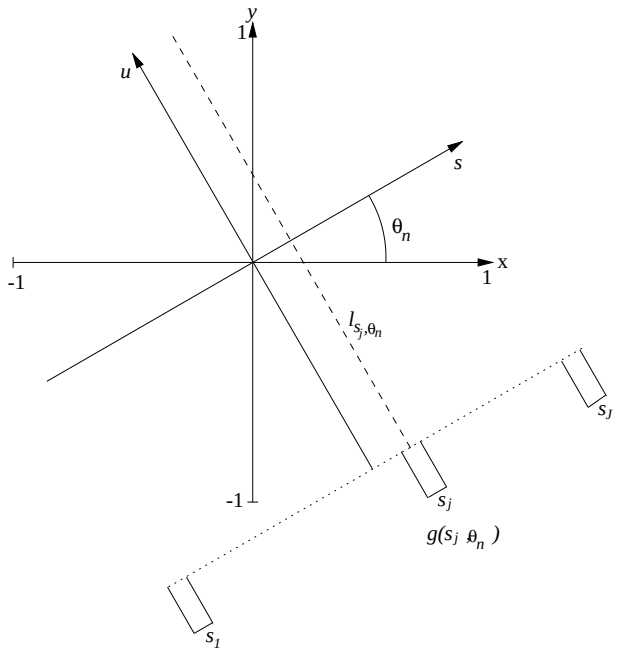


Figure 2.2: Detection with a finite number of views N and a finite number of detectors J.

version of the Radon inversion formula must be developed. This means that

the Fourier transforms must be discretized, and that the integral in the operator $\mathcal{B}$ in (2.13) must be approximated using summation and interpolation. Summarizing, a fast Fourier transform [20] is used for Fourier transformation of the discrete functions and the back-projection operator $\mathcal{B}$ of (2.13)is approximated by

$$[\mathcal{B}\tilde{g}](x,y) = \frac{\Delta}{2\pi}\sum_{n=0}^{N}\tilde{g}(x\cos n\Delta + y\sin n\Delta, n\Delta).$$

The $|S|$ of (2.5) is generalised to a function $F(S)$, which will be investigated in greater detail later. An interpolation operator $\mathcal{I}$ is used, to handle problems with our discrete data, and finally we apply the *Fourier Convolution theorem* [21]:

$$\mathcal{F}[f_1 * f_2] = \mathcal{F}f_1 \times \mathcal{F}f_2 \tag{2.20}$$

where $*$ denotes the convolution operator defined by

$$(f*g)(x) = \int f(x-y)g(y)dy.$$

The Fourier operators of (2.20) can be replaced with inverse Fourier operators, and multiplication $\times$ can be interchanged with convolution $*$. Including all these details, we get

$$f(x,y) = \mathcal{B}\mathcal{F}_1^{-1}\left([|S| \times \mathcal{F}_1 g(s,\theta)]\right) \approx \mathcal{B}\mathcal{I}(\mathcal{F}^{-1}F(S) * g(s,\theta)),$$

in agreement with [2] and [14]. If we denote the inverse, fast Fourier transform of $F(S)$ by q, we are able to reconstruct f by evaluating

$$f(x,y) = \mathcal{B}\mathcal{I}(q * g(s,\theta))(x,y). \tag{2.21}$$

The function $F(S)$, generalizing the behaviour of $|S|$, is called the *filter function*, and the reconstruction based on (2.21) is denoted *filtered back-projection*. The properties of different filter functions are demonstrated in section 2.3.1 below. Filtered back-projection is very suitable for computer implementation, and the data analysed in this thesis are based on exactly that reconstruction algorithm. Other reconstruction methods exist, among these *Algebraic Reconstruction Technique* (ART), in which the reconstuction is based on linear algebra, which gives rise to other problems than the ones discussed in this thesis. Suitable references on this topic are [7], [9], [10] and [13].

2.3.1 Filter functions

In this section, we will have a closer look at the filter function $F(S)$, introduced above.
Let us imagine that each projection $g(s,\theta)$ is a sum of a "true" signal $p(s,\theta)$ and a noise contributing term $\epsilon(s,\theta)$:

$$g(s,\theta) = p(s,\theta) + \epsilon(s,\theta)$$

The Fourier transform of g will give us a sum of the Fourier transforms of the signal and noise contributions. If we assume that ϵ is *white* noise with mean value 0 and variance σ^2, the Fourier transform of ϵ will also be white noise with mean value 0 and variance σ^2. Since the Fourier transform of p declines with increasing frequency, the signal to noise ratio will be lower at high frequencies than low frequencies, meaning that the Fourier transform of g is less trustworthy at high frequencies than at low. As earlier demonstrated, the Radon inversion formula (2.5) contains a $|S|$, which further amplifies the noise in g. To help suppress these effects, we replace $|S|$ by

$$F(S) = |S|W(S),$$

where $W(S)$ is a function which limits the used interval of frequencies, and modifies the shape of $|S|$. These functions are typically refered to as windows.

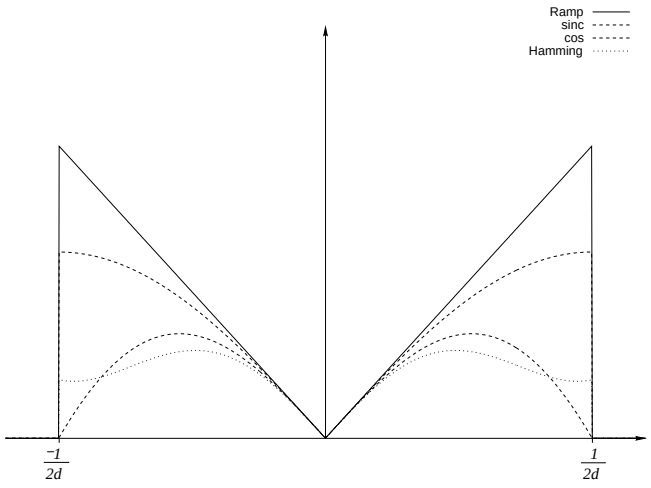


Figure 2.3: The four investigated filters.

Let us take a brief look at four commonly used windows:

a) Frequency limitation (Ramp):

$$W(S) = \Pi(Sd),$$

b) Low pass sinc:

$$W_{\text{sinc}}(S) = \text{sinc}(Sd)\Pi(Sd)$$

c) Low pass cos:

$$W_{\cos}(S) = \cos(\pi Sd)\Pi(Sd)$$

d) Hamming low pass window:

$$W_{\text{Hamming}}(S) = (\alpha + (1-\alpha)\cos(2\pi Sd))\Pi(Sd)$$

$\Pi(Sd)$ is a function with value 1 below and 0 above the frequency $Sd = \frac{1}{2}$. The four different types of filters are pictured in figure 2.3. The different types of filters have been developed for reduction of interference effects and enhancement of contrast in the final reconstructed image. A reference on this topic is [2].

Chapter 3

Scanner technology

The mathematics outlined in chapter 2 makes the construction of different types of scanners possible. In the following sections two main types of scanners are discussed. As both scanner types rely on the interaction of photons with matter, a section on this topic is also included.

3.1 Photon - matter interaction

The field quantum of electromagnetic radiation, also known as the photon, has in our field of application typically energies of around 10^4 – 10^5 eV. The interactions that take place between photons and matter[1] depend strongly on the energy of the photon and the composition of the matter, i.e. the binding energies of the electrons within the matter. As photons travel in matter, three types of interactions are important:

(*i*) Photo-electric effect. A photon delivers all of its energy to an atom which expells an electron.

(*ii*) Compton scattering. Elastic scattering between a photon and an electron.

(*iii*) Pair production. A photon of energy *at least* equal to 2 times the rest mass of a single electron disintegrates, exciting the electron- and positron field.

Since the typical energies in both transmission and emission tomography (explained in sections 3.2 and 3.3) are in the range of 10^4 – 10^5 eV, pair production can be excluded as source of attenuation. In materials with a

[1]Matter is in this sense meant as everything we would like to investigate.

low Z e.g. brain tissue, the attenuation due to the photo-electric effect can be excluded, since this contribution is small compared to that of Compton scattering (figure 3.1). Since only Compton scattering is important, we will further investigate this effect. For a more detailed calculation of all of the involved cross sections, see e.g. [4].

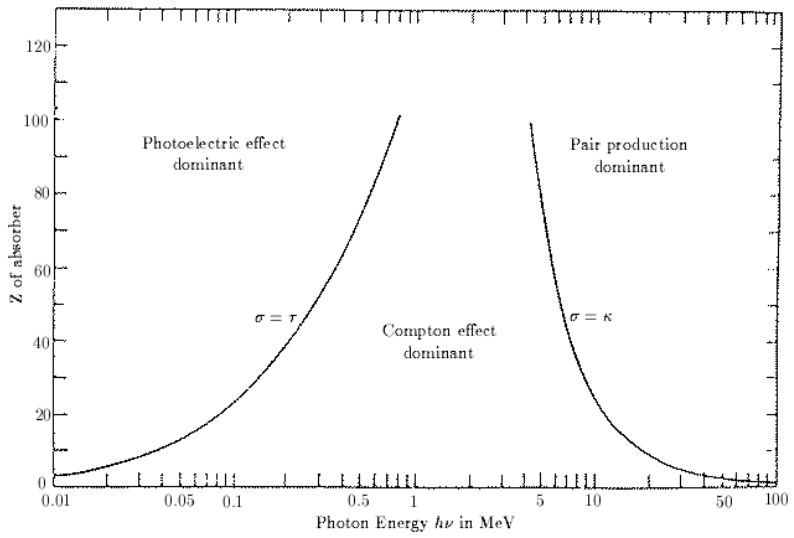


Figure 3.1: The most important contributions to the total cross section of absorption, reprint from [4].

3.1.1 Absorption due to Compton scattering

Let us remind ourselves of the *Klein-Nishina* cross section, to investigate the attenuation due to Compton scattering

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2}{2}\left(\frac{1}{1+\alpha(1-\cos\theta)}\right)^2\left(\frac{1}{1+\alpha(1-\cos\theta)} + \alpha(1-\cos\theta) + \cos^2\theta\right),$$

where $\alpha = \frac{h\nu_i}{mc^2}$, where ν_i is the frequency of the photon. The classical electron radius, or the Thomson length is $r_0 = 2.82 \cdot 10^{-15}$ m. For X-rays in general $\alpha < 1$, and for annihilation processes $e^+ + e^- \rightarrow \gamma_1 + \gamma_2$, $\alpha = 1$.
The total cross section is obtained by integration over all solid angles

$$\sigma = 2\pi r_0^2\left(\frac{1+\alpha}{\alpha^2}\left(\frac{2(1+\alpha)}{1+2\alpha} - \frac{1}{\alpha} + \ln(1+2\alpha)\right) + \frac{1}{2\alpha}\ln(1+2\alpha) - \frac{1+3\alpha}{(1+2\alpha)^2}\right).$$

Using this cross section, we are able to determine the probability that a photon is scattered travelling through matter of thickness $\Delta x = 1$ mm

$$P = \Delta x N Z \sigma_{ann} = 0.0096 \simeq 1\%$$

where $NZ = 3.35 \cdot 10^{29}$ m^{-3} is the density of electrons in water. In other words, 99% of a beam of photons will penetrate 1 mm of brain tissue, 37% will penetrate 10 cm. Hence, the attenuation is at a level which makes the use of photons with energies in the range of 10^4 – 10^5 eV possible for e.g. brain imaging, both when absorption is wanted and unwanted, as explained in the following sections.

3.2 Transmission tomography

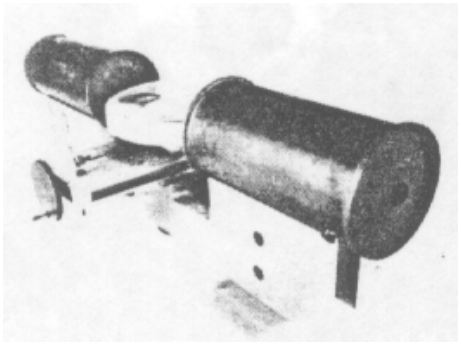


Figure 3.2: The first CT scanner used by Allan M. Cormack. Reprint from [3].

The first type of transmission tomography simply generalised the ordinary, planar X-ray absorption imaging to 3D. In planar X-ray imaging, X-rays are shined through an object, different parts of the object absorb X-rays to different extent, and hence a shadow image of the interior of the object is created. The technique named Computer (Aided) Tomography or simply CAT/CT, extract 3-dimensional information from a number of shadow images or projections of the absorbing parts of the object investigated.
The technique was independently developed in Great Britain and South Africa by physicists Godfrey N. Hounsfield and Allan M. Cormack. In 1979 they were adwarded the Nobel price in medicine for their discoveries. In the Nobel lecture [3] of Allan M. Cormack, one can read the exciting story of his invention. His work began in 1956 when he was became involved with radiotherapy treatment planning at the Groote Schuur Hospital of Cape Town.

Using the scanner depicted in figure 3.2 he investigated an object with a known inner structure, also called a phantom. Cormack was until 1970 unaware of the paper [1] by Johann Radon, so actually the first CT scanner was not directly based on Radon's work, but a less general method developed for the purpose by Cormack. Below, the physical foundation of the CT scanner is explained.
It is well known that the intensity of an X-ray beam travelling through a thickness x of a material with linear attenuation coefficient μ is

$$I(x) = I_0 e^{-\mu x},$$

where I_0 is the incident intensity of the beam. Knowing the incident intensity I_0, we are able to create projections[2] of the function $\mu(x, y)$, which maps the linear attenuation coefficient on a slice of the object investigated. The projections are simply taken to be

$$g(s, \theta) = \ln \frac{I_0}{I(s, \theta)}.$$

In figure 3.2 below, the technical design of the CT scanner is presented. In the first generation CT scanner (top, left), collimated radiation from the X-ray source and a single detector is used. Hence both translation and rotation of the detection system as well as the source is needed. Generation II (top, right) scanners uses a slightly less collimated beam, and a detector array. In generation III scanners (bottom, left) much less collimated radiation and a larger detector array is introduced, which removes translation of the detection system. In generation IV CT scanners (bottom, right), a stationary ring of detectors is used, only the source is rotated around the object. The advantage of using less collimated radiation is that a larger portion of the radiation emitted from the source is used for the imaging. Also the time of exposure is reduced, as a complete projection can be recorded at once, since the detectors need not be translated. The disadvantage of using non-collimated radiation is that a larger portion of the unwanted Compton scattered photons are recorded along with the transmitted ones, giving rise to a lower signal to noise ratio.

3.3 Emission tomography

In emission tomography the object investigated is also the source of radiation. An amount of radioactive material is inserted (in medicine by inhalation or

[2]Here, projections mean the type of mathematical projections defined in chapter 2.

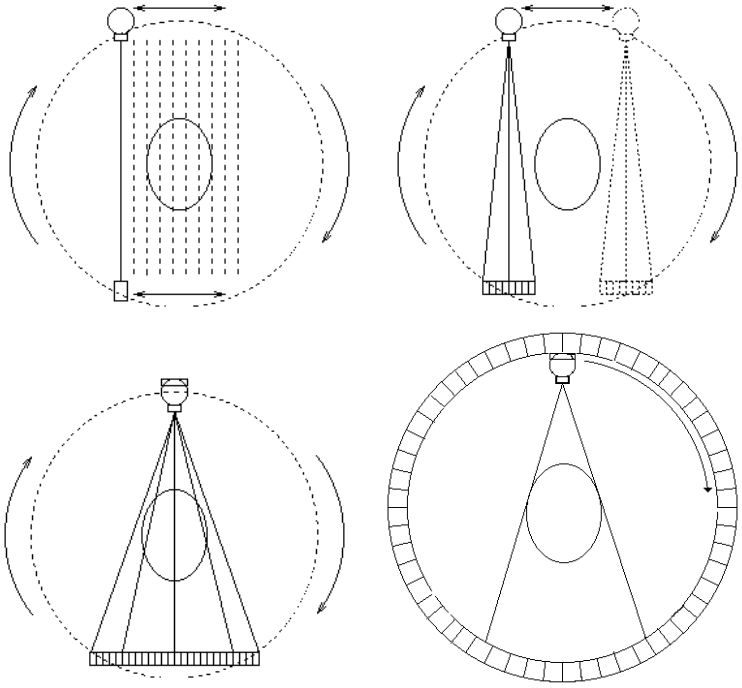


Figure 3.3: Development of the CT scanner. This figure is a redrawn version of a figure from [5].

injection) into the object and projections of the deposition of radioactive matter are recorded. In medical applications, emission tomography is used for diagnostics, studies on regional cerebral blood-flow (RCBF), metabolism and activation studies. Below, two examples of emission tomography are given.

3.3.1 SPECT

In Single Photon Computerized Tomography (SPECT), radioactive matter emitting single photons is inserted in the object. An example is $^{133}_{54}$Xe [4], a β^- radioactive isotope with a half-life of 5.27 days. The relaxation energy is 427 keV, and in almost all relaxations, the total energy of the expelled electron is 347 keV. In 36% of the relaxations, the rest of the relaxation energy is carried away with a photon, and in the other cases the left over energy is consumed during internal nuclear processes. The mentioned 36% of the events expel a photon of energy 80 keV, an energy in the X-ray range, suitable for tomography. As mentioned in section 3.1, the photons can be

either absorbed or transmitted interacting with the matter of the object. Of course only photons exhibiting the latter process are of interest for the tomography as we want to reconstruct the distribution of radioactive matter in the object.
To discriminate the scattered part of the photons from the wanted signal, a number of measures are taken. The detectors are set up to only detect photons with energy in the wanted range, that is $E \approx 80$ keV in the above example. We insert a collimator (see figure 3.4) to ensure that the detected

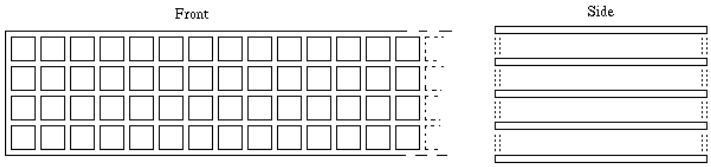


Figure 3.4: Collimator used in SPECT scanners.

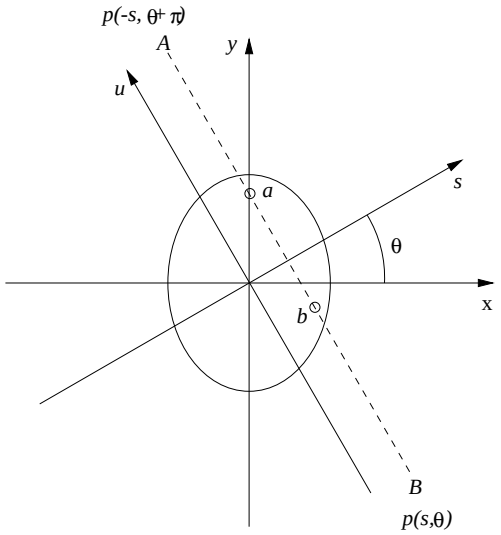


Figure 3.5: Projection summation in SPECT.

photons originate from a well defined part of the object investigated. Using a collimator, the solid angle of detection is reduced, hence the position of the decaying matter is less indefinite. Also a phantom is used to deduce a suitable correction to the measured signal.

The distance travelled by each photon within matter varies, hence the probability of absorption is different for each photon. Taking this into account, "opposite" projections are added - see figure 3.5. A photon emitted at a has a higher probability of being recorded at detector A than a photon emitted from b. To take this into account, signals detected at A and at B ($p(s,\theta)$ and $p(-s,\theta+\pi)$) are added.

3.3.2 PET

Positron Emission Tomography (PET) is an other example of emission tomography. In this case a positron emitting (β^+-radioactive) isotope is inserted into the object. In medical applications, usually a short-lived isotope with few neutrons is inhaled or injected. An example of such an isotope is

$$^{15}_{8}\mathrm{O} \rightarrow {}^{15}_{7}\mathrm{N} + e^+ + \nu \tag{3.1}$$

$$e^+ + e^- \rightarrow \gamma_1 + \gamma_2 \tag{3.2}$$

The positrons of process (3.1) are expelled with an initial speed which makes it unlikely that process (3.2) happens. So before entering the process (3.2), the positron is slowed down by scattering against electrons until it has a speed which is low enough to make (3.2) happen. The fact that (3.2) only

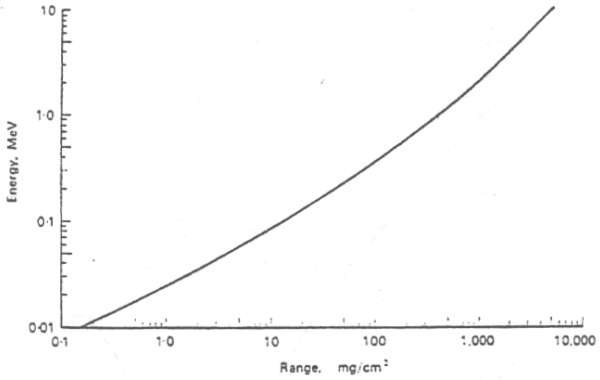


Figure 3.6: Range-Energy curve for β-particles, reprint from [4].

happens at low speeds, reflect that the cross-section of this process is an overlap integral of the wavefunctions of the electron and the positron in an interaction potential. When the positron has lower speed it is less localised, making the value of the overlap integral higher. As the positron slows down, it travels a distance from the originating point. This gives rise to a point

spread function apart from the one reflecting the discrete sampling situation depicted in figure 2.2. The positron is emitted with a typical kinetic energy of 1.7 MeV, which according to figure 3.6 [3] corresponds to a maximum range of about 500 mg/cm^2. In human tissue (which is mostly water of density 1 g/cm^3) this corresponds to a maximum travelled distance of 5 mm.
In process (3.2), the positron interacts with an electron of the surrounding matter, and an annihilation process takes place. Hence two photons of equal energy (511 keV) are expelled in (almost) opposite directions, due to conservation of total momentum. Since each annihilation produces *two* photons of equal energy we only want to record those photons detected (practically) simultaneously and in pairs, single photons are considered noise. Present day electronics is not fast enough to measure the maximal time interval between the two photons, as this is of order $t = \frac{1\ \mathrm{m}}{c} \approx 3$ ns. When electronics become that fast, the measured difference of time can be utilised for determination of the point of decay of the positron. The PET detection principle is illustrated in figure 3.7. The result of each successful measurement is that

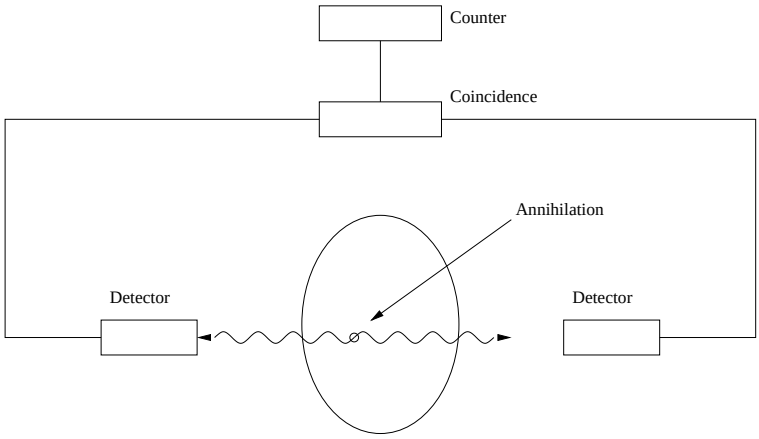


Figure 3.7: Simultaneous detection of two photons in PET.

an annihilation took place somewhere on the line between the two detectors involved in the detection. Hence, only annihilations taking place within the slice being investigated with photons moving within the detection plane are recorded. In principle a recorded event could be the result of two simultaneous one-photon decays. A correction for this effect is done by investigating a phantom.
If the positron of process (3.2) has a certain, small amount of kinetic energy when interacting with the electron, the angle between the trajectories of the

[3] The range-energy relationship of positrons does not differ much from that of electrons.

two expelled photons will differ slightly from π. Experimental surveys [11], [12] conclude that a good approximation to the deviation from π is a Gaussian distribution $e^{-\lambda\theta^2}$, shown in figure 3.8. In this figure it is seen that the maximum deviation $\Delta\theta$ from π is about $\frac{1}{100}^{\mathrm{O}}$. The maximum angular devi-

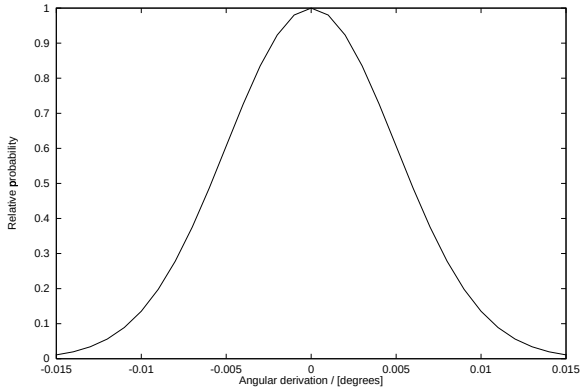


Figure 3.8: Distribution of photon angle difference from π in positron-electron annihilation. Reprinted from [13].

ation can be used to approximate the uncertainty of the position where the annihilation took place. Looking at figure 3.9, for a PET scanner of diameter 1m, a deviation $\Delta\theta = \frac{1}{100}^{\mathrm{O}}$ corresponds to a $\Delta s = 0.5\ \mathrm{m} \sin \frac{1}{100}^{o} \approx 0.1$ mm, which in the worst case means that the event is recorded in a neighbour detector. With a cirular detector geometry as in figure 3.9, this means that the two photons are recorded as if the annihiltion took place at position a distance $d/2$ away from the true point of annihilation, where d is the width of the detectors. In the General Electric Advance scanner (figure 3.10) of the Rigshospitalet PET unit, the detection rings are of diameter 92.7 cm with 672 detectors in each ring. This makes $d = \frac{0.972\ \mathrm{m}}{2} \sin \frac{2\pi}{672} \approx 4$ mm, meaning an uncertainty of 2 mm in determination of the annihilation point. Including this uncertainty as well as the uncertainty from the travelling of the positron, the overall uncertainty in detection of the originating point of the positron is $\Delta r = \sqrt{(5\ \mathrm{mm})^2 + (2\ \mathrm{mm})^2} \approx 5.4$ mm.
Scanner parameters such as the number of detectors, should not be chosen for a better theoretical resolution than the position uncertainties dictated by the physical situation.

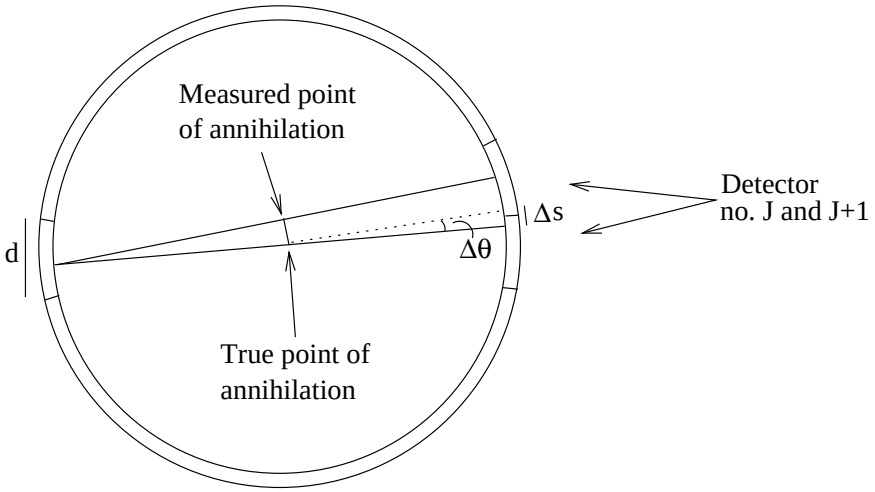


Figure 3.9: θ-deviation influence on detection position. Lengths are not mutually comparable.



Figure 3.10: The GE Advance scanner, Courtesy of General Electic Corp. Image from the PET unit homepage `http://www.pet.rh.dk/uk/scanners.html`.

Chapter 4

Point source reconstruction

As mentioned in the preface, the goal of this thesis is to quantify how a point source is reconstructed through *filtered back-projection* (FBP), as function of different parameters influencing the reconstruction. This chapter includes firstly a section of theoretical considerations on point source reconstruction from [15] by Stig Steenstrup. Secondly a section on considerations regaring simulation of signals from a point source and finally a section describing the features of the developed simulation routine. Before proceeding, the reader is reminded of the detection situation, depicted in figure 2.2.

4.1 Theoretical properties of a point source

The content of this section summarizes the results found in [15].

The starting point of the paper is an expansion of the function $f(r,\phi)$ on the orthogonal set of Zernike[1] polynomials:

$$f(r,\phi) = \sum_{n=-\infty}^{\infty} \sum_{l=0}^{\infty} f_n^l R_{|n|}^l(r) e^{in\phi} \tag{4.1}$$

From the expansion and formulas from appendix A, an expression for $g(s,\theta)$ is achieved

$$g(s,\theta) = \sum_{n=-\infty}^{\infty} \sum_{l=0}^{\infty} \frac{2f_n^l}{|n|+2l+1} \sqrt{1-s^2}\, U_{|n|+2l}(s)\, e^{in\theta}, \tag{4.2}$$

[1]The properties of these polynomials can be found in appendix A, which is a reprinted version of an appendix in [15].

where $U_n(x)$ is a Chebyshev polynomial of the second kind. An explicit inversion formula is obtained using the orthogonality of the exponentials and of the Chebyshev polynomials:

$$f_n^l = \frac{|n|+2l+1}{2\pi^2} \int_{-1}^{1} ds \int_0^{2\pi} d\theta \; g(s,\theta)\, U_{|n|+2l}(s)\, e^{-in\theta} \tag{4.3}$$

Using equation (4.2) and expansion in the Chebyshev polynomials, a set of discrete samplings of $g(s,\theta)$ is achieved

$$\breve{g}_n^l = \sum_{k=0}^{J-1} w_k \breve{g}_n(\alpha_k)/\sqrt{1-\alpha_k^2}\; U_{n+2l}(\alpha_k), \tag{4.4}$$

where w_k are $k-$dependent weights and α_k are the roots of $U_J(x)$, J being the number of s-samplings. $\breve{g}_n$ is defined by

$$\breve{g}_n(s) = \sum_{\nu=0}^{N-1} g(s,\theta_\nu) \exp\left(-2\pi i n\nu/N\right), \tag{4.5}$$

sampling the projections $g(s,\theta)$ at N values of θ. Thus, in (4.4) a discrete set of projections with N values of θ and J values of s has been given. With f being a point source $f(x,y) = \delta(x-x_0)\delta(y-y_0)$, $g(s,\theta)$ becomes

$$g(s,\theta) = \delta(s - r_0\cos(\theta-\phi_0)), \tag{4.6}$$

with $x_0 = r_0\cos(\phi_0)$ and $y_0 = r_0\sin(\phi_0)$. Inserting this result in (4.4) and reconstructing using (4.3), a reconstruction of the point source using only a discrete set of samplings is obtained

$$G(\mathbf{r},\mathbf{r}_0) = \sum_{n=-(N-1)}^{N-1} \sum_{l=0}^{J-1} \frac{|n|+2l+1}{\pi}\; R_{|n|}^l(r_0)\, R_{|n|}^l(r)\, e^{in(\phi-\phi_0)}. \tag{4.7}$$

Using (A.5), $G(\mathbf{r},\mathbf{r}_0)$ is written:

$$G(\mathbf{r},\mathbf{r}_0) = \tag{4.8}$$
$$\frac{2}{\pi}{\sum_{n=0}^{N-1}}' \; \frac{J(J+n)}{2J+n} \frac{R_n^J(r_0)R_n^{J-1}(r) - R_n^J(r)R_n^{J-1}(r_0)}{r_0^2 - r^2}\; \cos(n(\phi-\phi_0)),$$

where the prime on the summation indicates that the term for $n=0$ has to be halved. The function $\bar{f}(r,\phi)$ reconstructed by

$$\bar{f}(r,\phi) = \sum_{n=-(N-1)}^{N-1} \sum_{l=0}^{J-1} f_n^l \; R_{|n|}^l(r)\, e^{in\phi}, \tag{4.9}$$

with $f_{-n}^l = f_n^{l*}$ can also be written

$$\bar{f}(r,\phi) = \int_0^1 r'dr' \int_0^{2\pi} d\phi' G(\mathbf{r},\mathbf{r}') f(r',\phi'). \tag{4.10}$$

Since the Greens function or point-spread function in (4.7) or (4.8) connects the original $f(r,\phi)$ (in this case a δ-function) with the reconstruction $\bar{f}(r,\phi)$, $G(\mathbf{r},\mathbf{r}')$ contains the full information about the reconstructed resolution. Thus, G is the theoretical point-spread function to be compared with actual reconstructed data. Steenstrup [15] claims that $G(\mathbf{r},\mathbf{r}')$ is the *best possible* point-spread function, hence has the narrowest possible width, using J detectors at N angular views (figure 2.2).
The paper [15] by Stig Steenstrup also includes a set of figures showing properties of G from (4.7) for different values of the number of θ-samplings N and s-samplings J. The figures are reprinted below (figure 4.2–4.6).
In figure 4.2, $G(\mathbf{r},\mathbf{r}')$ is depicted as function of r, for different values of N and J, the point source positioned at $(r_0,\theta_0) = (0,0)$. Figure 4.3 is similar to figure 4.2, except that the point source is placed at $(r_0,\theta_0) = (0.8,0)$. In figure 4.4 $G(\mathbf{r},\mathbf{r}')$ is viewed as function of θ, with $-\pi \le \theta \le \pi$, the point source positioned at $(r_0,\theta_0) = (0.8,0)$. In figures 4.5 and 4.6, the radial FWHM W_r and the transverse FWHM W_t (see Figure 4.1) of $G(\mathbf{r},\mathbf{r}')$ are viewed as functions of the point source position, r_0. The presented figures are the main target of the later comparison with reconstructed data from the author's reconstruction routine.

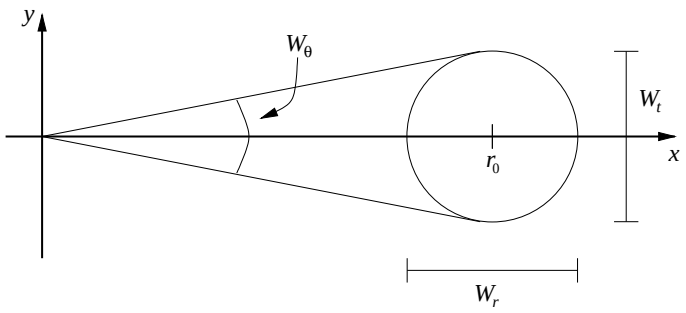


Figure 4.1: Point-spread function with radial width W_r, angular width W_θ and transverse width W_t positioned at $(r_0,0)$.

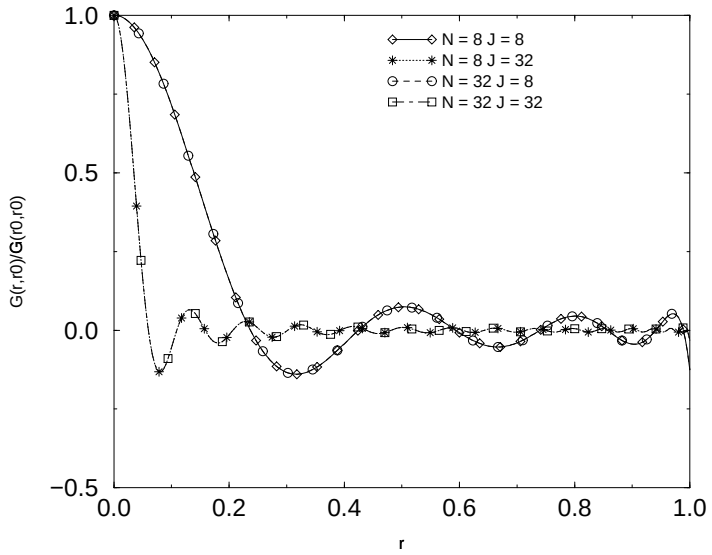


Figure 4.2: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0$.

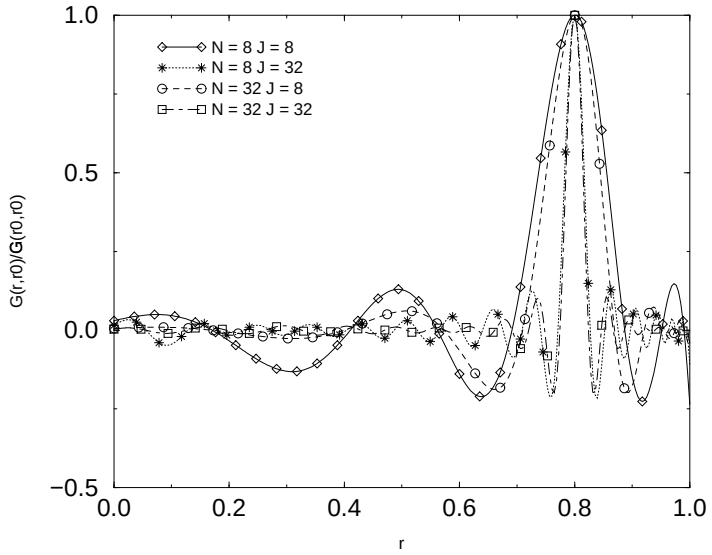


Figure 4.3: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0.8$.

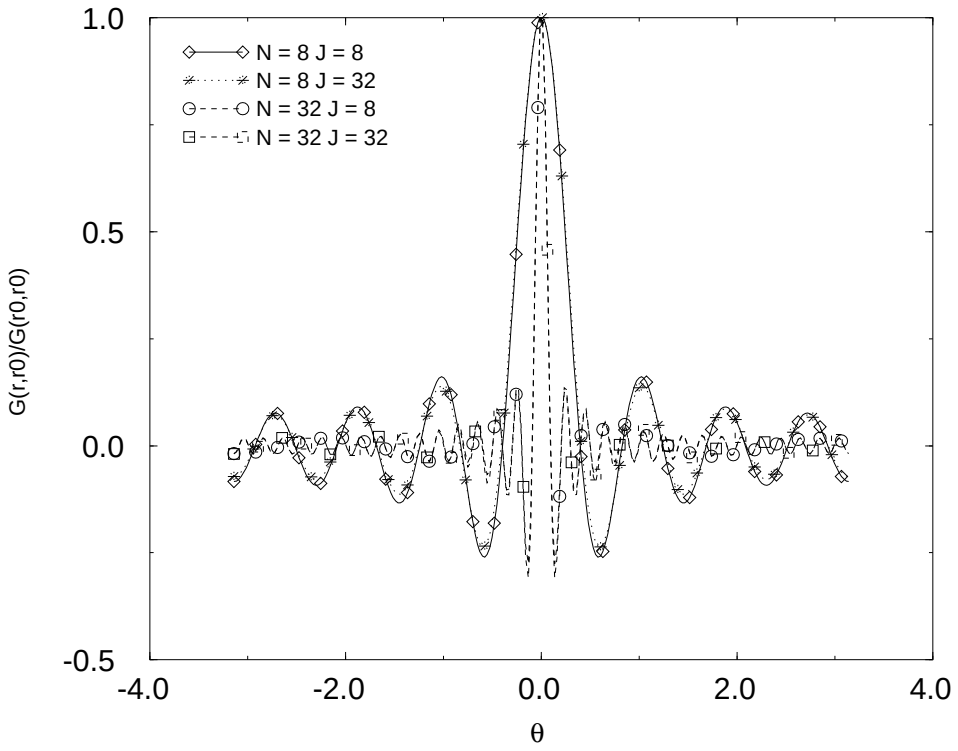


Figure 4.4: $G(r, r_0)$ as fct. of θ, point source positioned at $r_0 = 0.8$.

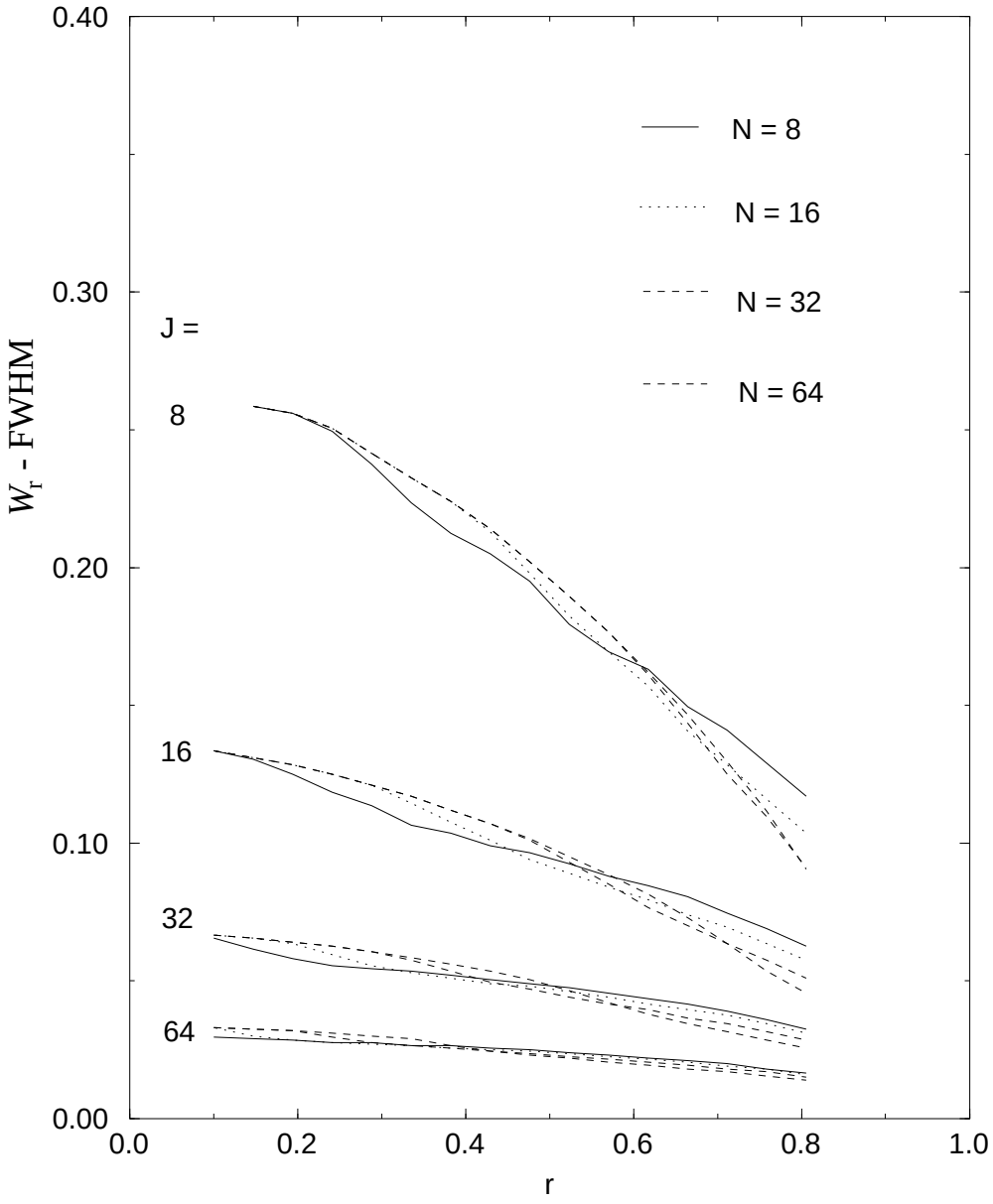


Figure 4.5: W_r of $G(r, r_0)$ as fct. of r_0.

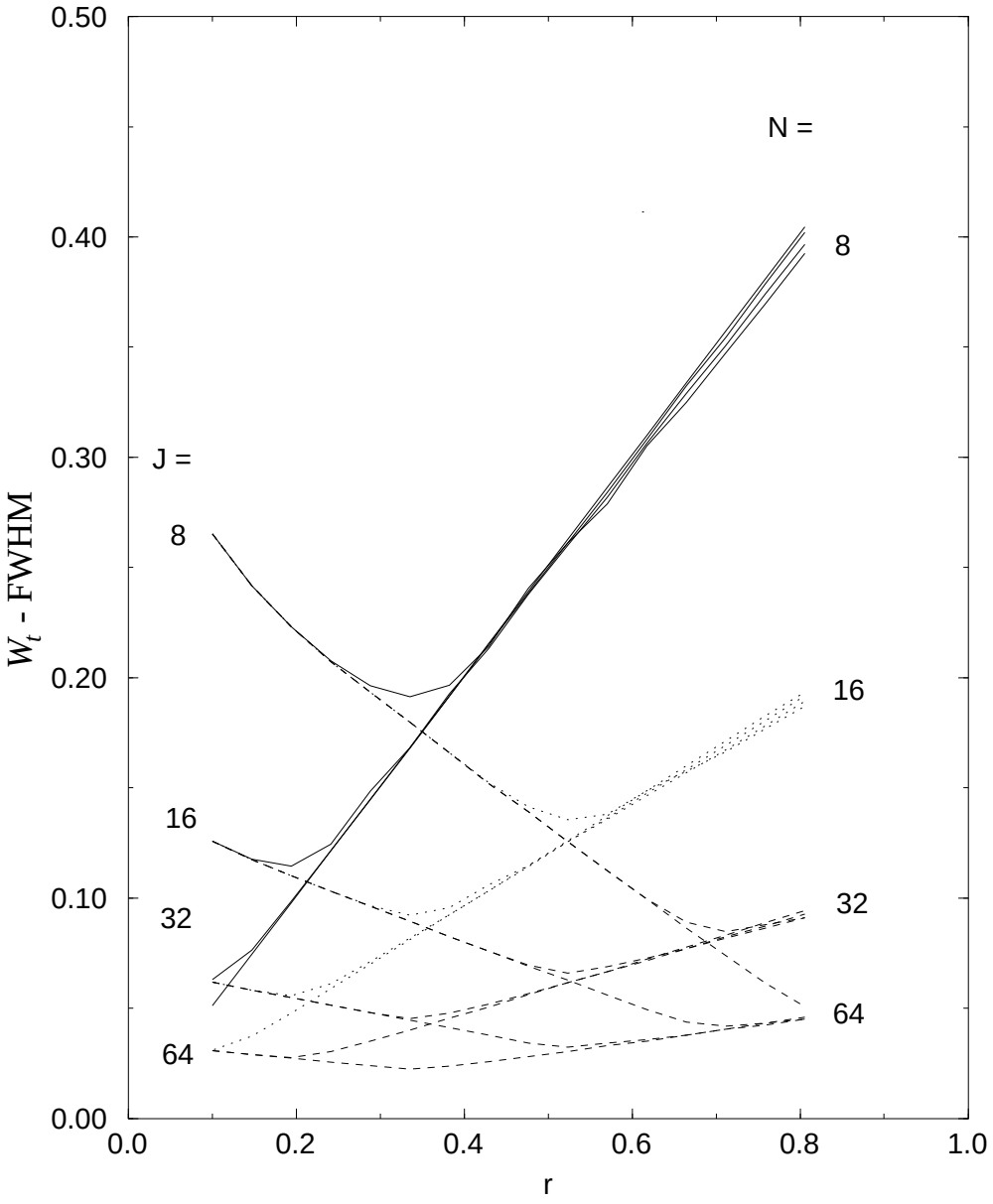


Figure 4.6: W_t of $G(r, r_0)$ as fct. of r_0.

4.2 Signal detection

4.2.1 Projections

In creation of the simulation routine the clean mathematical problem outlined in chapter 2 must clearly be simplified and assumptions regarding the signal detection must be made. In this section, the used assumptions and simplifications regarding projections of the point source are discussed. Naturally, a point source, descibed by a Dirac delta-function $\delta(x-x_0)\delta(y-y_0)$ of two variables x and y, is difficult to implement in the discrete framework of computers. Hence, our point source is approximated using a narrow Gaussian bell

$$f(x,y) = \exp\left\{-\frac{(x-x_0)^2}{2\sigma^2} - \frac{(y-y_0)^2}{2\sigma^2}\right\}, \tag{4.11}$$

which is only a good approximation, when $\sigma \ll d$, where $d = \frac{2}{J-1}$ is the width of the used detector.
When sampling projections of f we must, following (2.1) and figure 2.2, evaluate the set of integrals

$$g(s,\theta) = \int_{l_{s,\theta}} f(x,y)dl.$$

This formula does not account for the width and efficiency of the used detector, or in other words the efficiency curve of the detector. Two possible

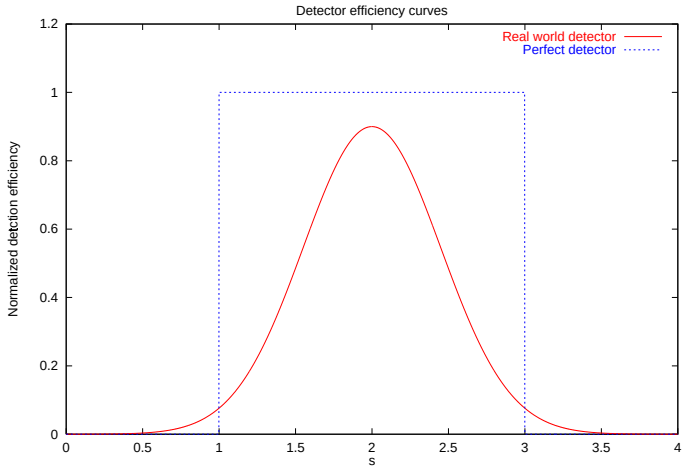


Figure 4.7: Efficiency curves of real world and perfect detector.

detector efficiency curves are shown in figure 4.7. Our detectors are assumed

to have a perfect, box-shaped efficiency curve, meaning an s-integration of the above projections. Implementing this feature, the new set of projections become

$$\tilde{g}(s,\theta) = \int_{s-\frac{d}{2}}^{s+\frac{d}{2}} g(s,\theta)ds = \int_{s-\frac{d}{2}}^{s+\frac{d}{2}} \left(\int_{l_{s,\theta}} f(x,y)dl \right) ds. \tag{4.12}$$

With a more complicated type of efficiency curve, a convolution should replace the s-integration in (4.12). In the chosen geometry (4.12) really means

$$\tilde{g}(s,\theta) = \int_{s-\frac{d}{2}}^{s+\frac{d}{2}} \left(\int_{-\infty}^{\infty} f(s\cos\theta - u\sin\theta, s\sin\theta + u\cos\theta)du \right) ds.$$

Using the result of section 2.2.2, the innermost integral has the form:

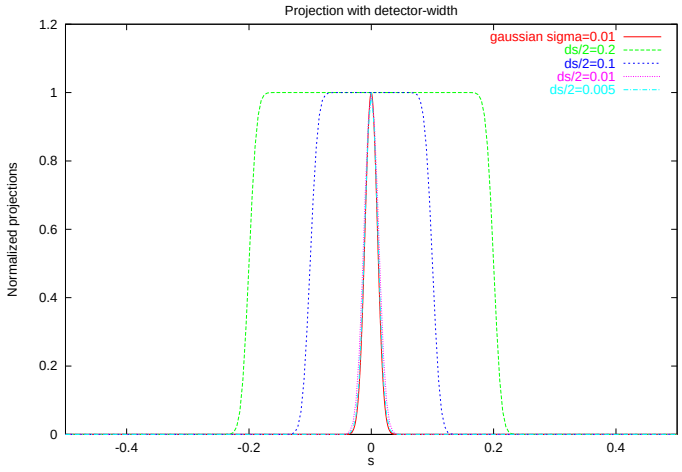


Figure 4.8: Projections accounting for detector width.

$$g(s,\theta) = \sqrt{2\pi}\sigma \exp\left\{ -\frac{(r_0\cos(\theta-\theta_0)-s)^2}{2\sigma^2} \right\},$$

so that the projections are achieved by evaluating

$$\tilde{g}(s,\theta) = \sqrt{2\pi}\sigma \int_{s-\frac{d}{2}}^{s+\frac{d}{2}} \exp\left\{ -\frac{(r_0\cos(\theta-\theta_0)-s)^2}{2\sigma^2} \right\} ds.$$

Using *Mathematica* and [23], I found that this integral becomes

$$\tilde{g}(s,\theta) = \pi\sigma^2 \left\{ \operatorname{erf}\left(\frac{s+\frac{ds}{2} - r_0\cos\theta - \theta_0}{\sqrt{2}\sigma} \right) - \operatorname{erf}\left(\frac{s-\frac{ds}{2} - r_0\cos\theta - \theta_0}{\sqrt{2}\sigma} \right) \right\}, \tag{4.13}$$

were $\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du$. The result looks quite as expected, the projections (4.13) are broadened with respect to the simple line-integrals of the Gaussian, as shown in figure 4.8. The projections implemented in the author's FBP program are of the form (4.13).

4.2.2 Noise

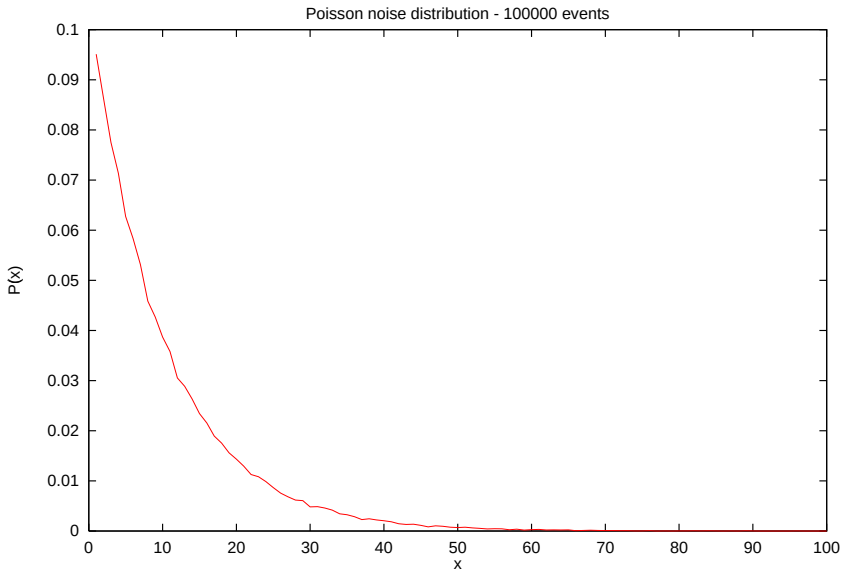


Figure 4.9: The background noise-distribution at 100000 events.

As noted in section 2.3, in the real world the detected signals do not behave like nice, analytical functions. Noise is present, and that fact has also been taken into account in the creation of the FBP routine. Two types of noise have been implemented, a background contribution based on the Poisson distribution, as well as a Gaussian distribution simulating errors due to detection statistics. The background contribution is implemented by applying $-\log(p)$ to a usual square-shaped distribution of numbers p between 0 and 1. The result is depicted in figure 4.9. The Gaussian distribution is made using $\frac{1}{N} \sum_{j=0}^{N} p_j - \frac{N}{2}$, where N is a large number, ideally ∞. In this case $N = 12$ seems to be a good approximation to ∞, as documented by the gaussian distribution of figure 4.10.

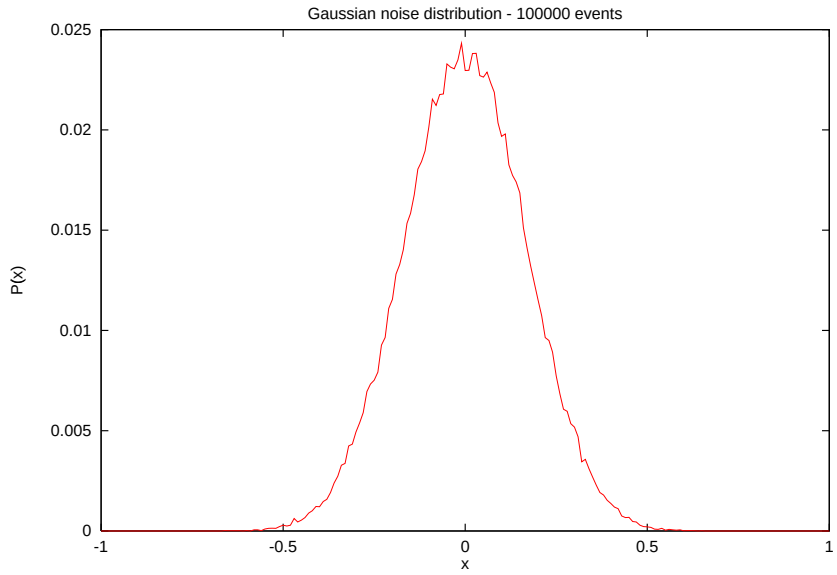


Figure 4.10: The Gaussian noise-distribution at 100000 events.

4.3 Implementation of simulation software

This section is an overview of the features of the developed FBP simulation software. Source code for compilation on most UNIX systems is available at `http://fys.ku.dk/~willend` and has also been placed on the enclosed cdrom. For use of the cdrom, see appendix C.

4.3.1 Schematic overview

As outlined in chapter 2, in particular section 2.3, a computer program based on FBP should include the following routines:

1. A routine for initialisation and loading the projections $g(s, \theta)$.

2. A convolution routine for convoluting $g(s, \theta)$ with $q = \mathcal{F}^{-1}F(S)$. For the best possible performance convolution is implemented using fast Fourier transforms.

3. A back-projection routine implementing the back-projection operator $\mathcal{B}$ from (2.13).

The following sections describe the mentioned parts of the reconstruction program.

4.3.2 Initialisation

Since the target of the work presented in this thesis is to quantify how a point source is reconstructed through FBP, a number of parameters characterising the point source must be specified. The parameters are

- (x_0, y_0) - position of the Gaussian "point" source.
- σ - width of the "point" source.
- J - the number of detectors.
- N - the number of angular projections.
- R_{SN} - signal to noise ratio.
- j_f - the filter[2] to use for reconstruction.

Furthermore, the user can specify whether errors should be added and what file names should be, etc. It is assumed, that the numbers of reconstruction sites (n_x, n_y) should both be comparable to $2J$. This value is chosen to make (n_x, n_y) match the resolution dictated by the detector number J, but still so low that the linear interpolation involved in the reconstruction does not influence badly. The initialisation part of the reconstruction program keeps track of parameters and calculates the projections $g(s, \theta)$ according to (4.13). Errors are added according to section 4.2.2, and the signal is scaled so that the peak value of the signal is approximately R_{SN} times the mean value of the background noise.

4.3.3 Convolution

In this part of the program the Fourier coefficients q of the chosen reconstruction filter are created analytically, following the guidelines in [2]. A convolution of these with the projections (4.13) is carried out using a convolution routine based on fast Fourier transformations, in agreement with the Convolution Theorem (2.20).

4.3.4 Back-projection

In the third and final part of the program, the back-projection operator $\mathcal{B}$ from (2.13) is used to reconstruct the defining function f of the point source on a grid of size (n_x, n_y). During these calculations, interpolation in

[2]One of the filters presented in section 2.3.1.

the discrete dataset is needed. In the program a simple linear interpolation method has been implemented. After completion of the reconstruction, f is re-normalized to unity, for easy comparison of different reconstructions and with the analytical results of [15].

4.3.5 Other features

In addition to the above mentioned features of the program, the following functionality has been implemented:

- The FWHM widths W_r, W_θ and W_t of the reconstructed function are extracted by simple linear interpolation in the reconstructed grid.
- Reconstructed data are saved in a consistent file format and log files are written for easy debugging.
- The described program takes parameters as command line arguments for easy use in parameter-varying scripts.
- Scripts have been written for Octave - a free implementation of the Matlab language. A number of scripts for variation of all the relevant parameters have been written. Data-analysis scripts have been written, for easy mesh- and contour-plots of reconstructed point sources, as well as 2D-plotting of the recorded point spread function widths.

Chapter 5

Simulation Data

The sections of this chapter describes the point source studies made with the author's computer program and the data from these investigations. Also, a section regarding the analysis of the data is provided.

5.1 Outline of the studies

As stated in the introduction, one of the main goals of this thesis is to investigate the validity of the theory presented in section 4.1. Hence, using the developed computer program, a series of figures similar to figures 4.2–4.6 should be produced. This type of study should be carried out for each of the filters presented in section 2.3.1. Possibly the influence of noise on the reconstruction should be investigated. Summarising, the set of investigations should for each filter, with and without noise include

- Variation of the number N of θ-steps as well as the number of detectors J. These variations should include at least the same range of parameters as used in [15] - figures 4.2-4.6.
- A variation of the position r_0 of the point source.
- For each value of r_0, a measurement of the widths W_r and W_t should be done, to investigate if the dependence of N and J on these is similar to that presented in [15] - figures 4.5–4.6.

In all of these cases the reconstruction should be done on a grid of n_x by n_y points, where both are approximately $2J$, of reasons argued in section 4.3.2. The angular position θ_0 of the point source will not be varied because of circular symmetry. In addition to these investigations, examples of variation of the point source width σ and the signal to noise ration R_{SN} should also be

made. If possible a general evaluation of the strength of each filter for use in the simulation study should be considered. All point source reconstructions have been carried out with a point source width of $\sigma = 10^{-50}$, which to good approximation is seen as a point source for any of the used values of J, N.

5.2 Analysis of reconstruction data

In agreement with the last section, we will now present the data achieved using the simulation routine, corresponding to the figures 4.2-4.6. Since the complete set of data is fairly large, only a subset of the data are presented within the thesis text. This section only includes data reconstructed using the Ramp filter, since the general conclusions made are similar for all filters. Data from the other filters are presented in appendix B, and a more complete set of data is presented in `HTML` pages on the enclosed cdrom. On this cdrom, the mentioned variations of σ and R_{SN} as well as r_0 variations can be found, visualised in animations. For use of the cdrom, consult appendix C. The notation $\{J, N\} = \{A, B\}$ in the figure titles means that the depicted function is shown for all values $A \leq J \leq B$ *and* all values $A \leq N \leq B$. The position unit $0 \leq r \leq 1$ on the shown figures is measured in the units dictated by the detection system, shown in figure 2.2, and various widths are measured in the same unit. Note also that the influence of noise on the widths W_r, W_t and W_θ (figure 4.1) is so large (figure 5.1), that only the error-free versions are presented. Unfortunately time did not allow further investigation into the influence of errors on the reconstructed widths, and as later stated, this could be a subject of further investigation.

5.2.1 Ramp filter

Looking at figure 5.2, the general shape of the point-spread function is similar to that of figure 4.2, including the small features found symmetrically around the main peak at $r_0 = 0$. These features will be referred to as higher order peaks. The two figures differ in the respect that the width of the main peak and the higher order peaks seem about a factor of 2 larger in figure 5.2 than in figure 4.2. Notice how the width of the main peak depends mainly on J, and that the shape of the higher order peaks depend mainly on N, which is not seen in figure 4.2. The shape of the higher order peaks also depend on θ, as shown in figure 5.4 and 5.10. Using the same set of parameters with errors included, the result depicted in figure 5.3 is found. Adding errors to some degree removes the N-dependence on the shape of higher order peaks. Results are similar using the other filters, as documented by appendix B.

The shape of higher order peaks is found to depend on the used filter, in agreement with the remarks on interference and contrast of section 2.3.1.

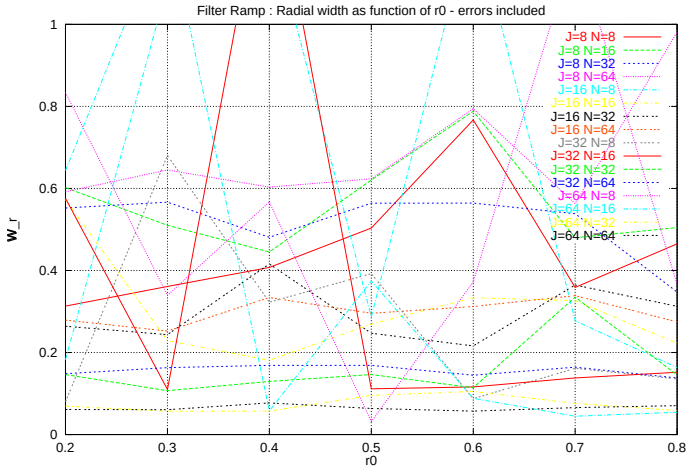


Figure 5.1: Noise influences badly on width of reconstruction.

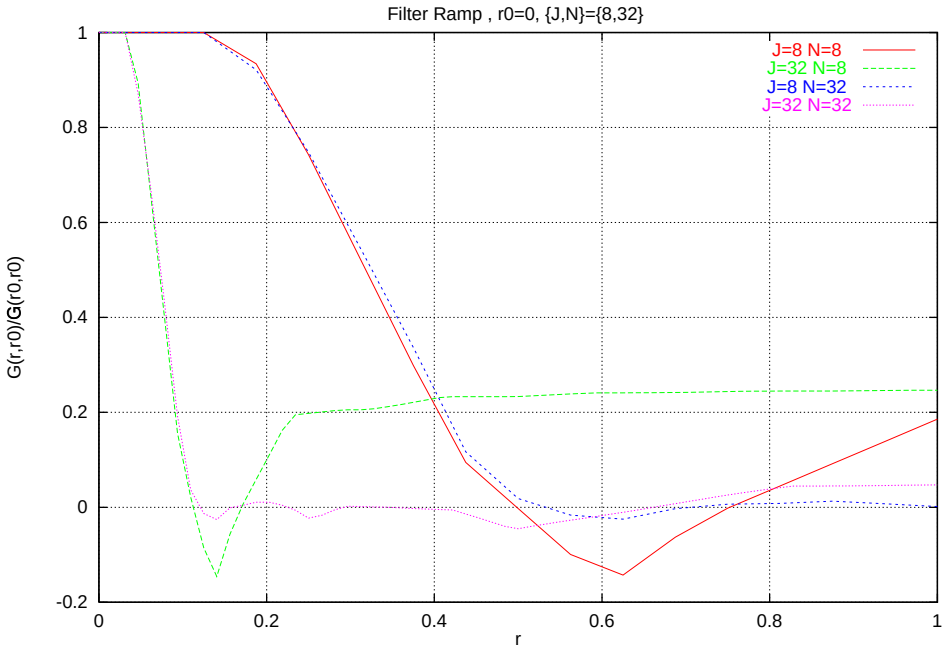


Figure 5.2: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0$.

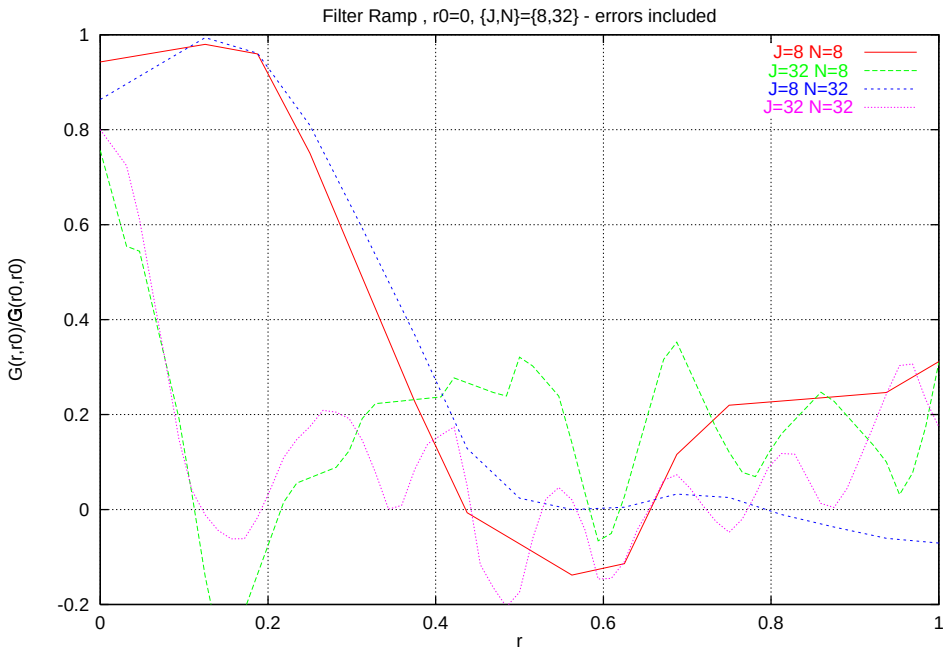


Figure 5.3: As figure 5.2, but errors added.

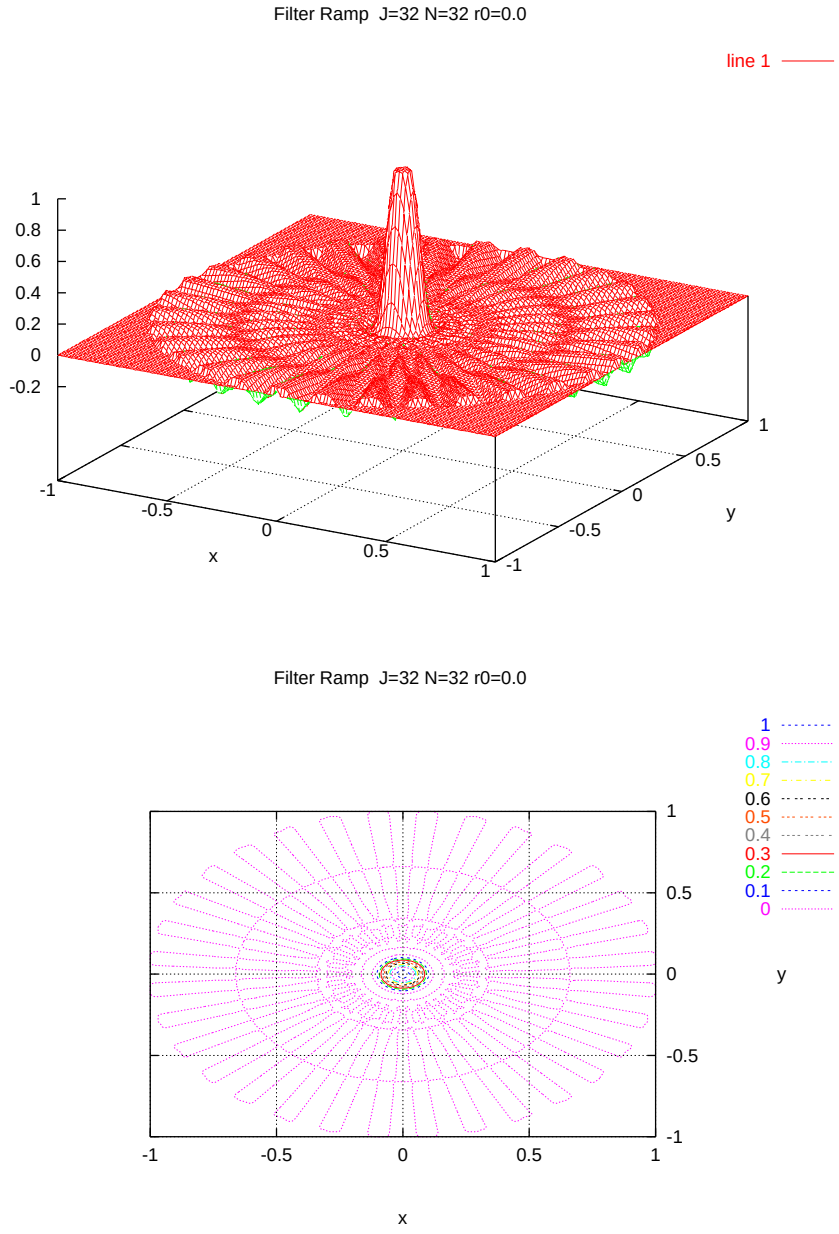


Figure 5.4: Contour and mesh plots, $N = 32, J = 32, r_0 = 0$, no error.

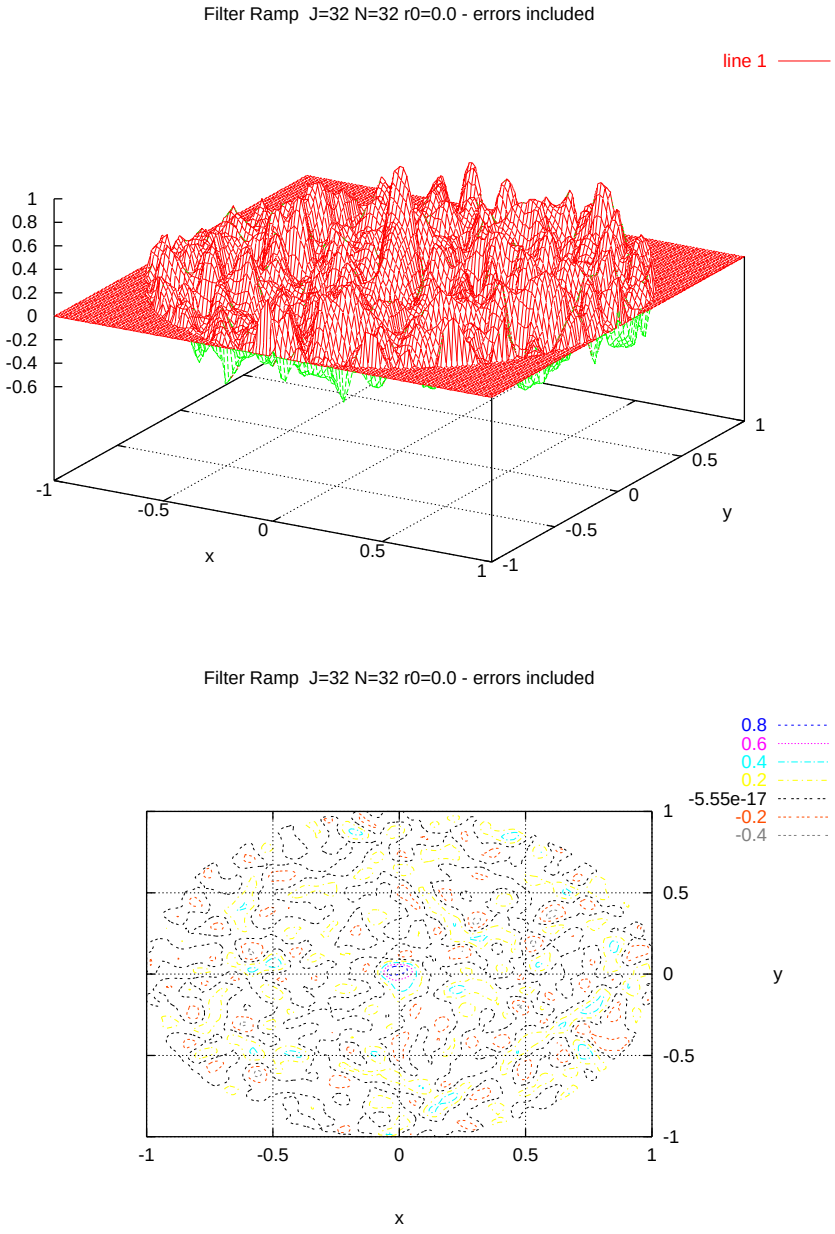


Figure 5.5: Contour and mesh plots, $N = 32, J = 32, r_0 = 0$, errors added.

In figure 5.6, the general shape of the point-spread function is again similar to that of figure 4.3, including the higher order peaks, but in this case, the width of the peaks is much larger in figure 5.2 than in figure 4.3. As in figure 5.2, the width of the main peak depends mainly on J, and the shape of the higher order peaks depend mainly on N, in this case in better agreement with figure 4.3. The shape of the higher order peaks also depend on θ, as shown in figure 5.8 and figure 5.10. Using the same set of parameters with errors included, the result depicted in figure 5.7 is found. Again errors influence the shape of the higher order peaks, as well as shift the center position of the main peak. In the cases where J is low, this could partly be due to weak statstics.
Results are similar using the other filters, as documented by appendix B.

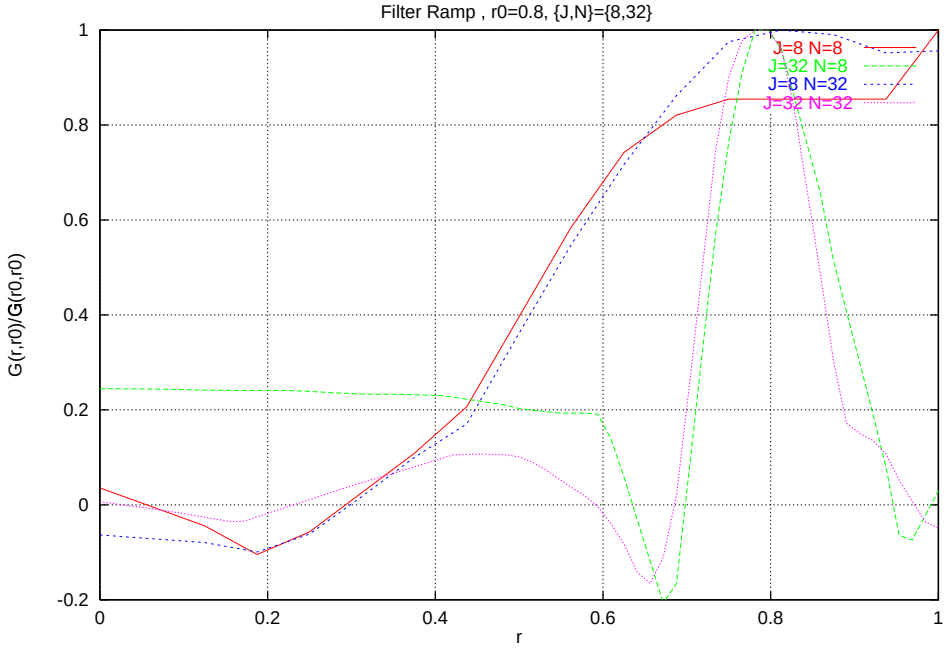


Figure 5.6: $G(r, r_0)$ as fct. of r, pointsource positioned at $r_0 = 0.8$.

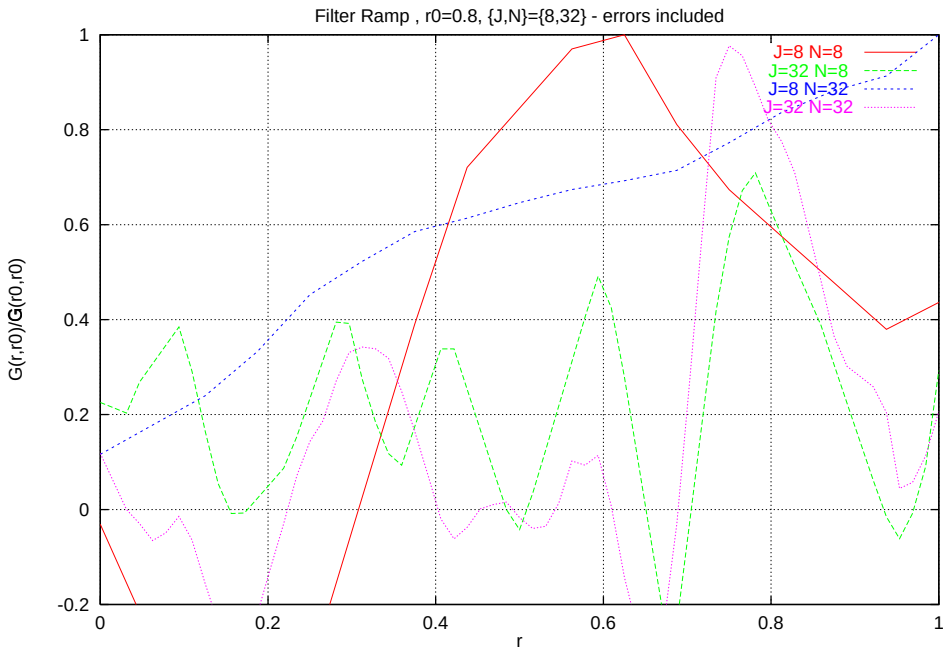


Figure 5.7: As figure 5.6, but errors added.

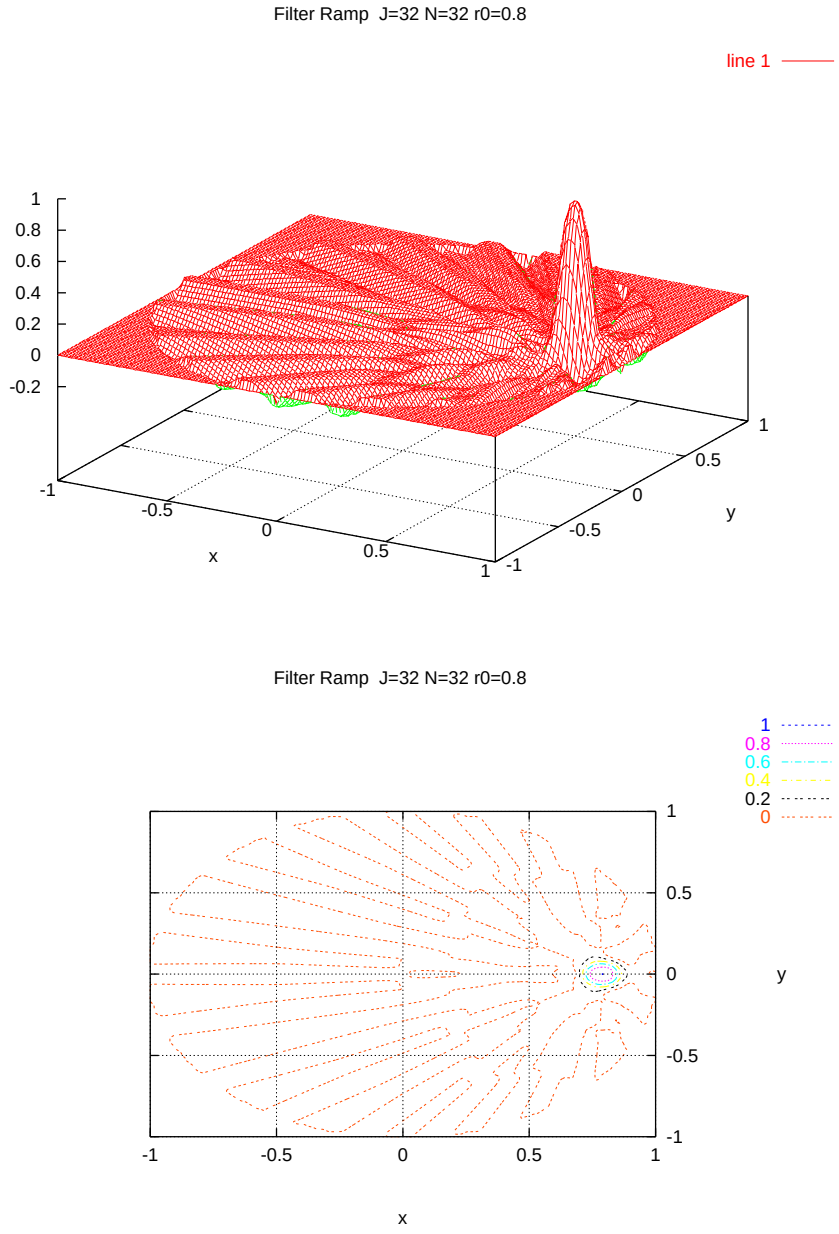


Figure 5.8: Contour plot, $N = 32, J = 32, r_0 = 0$ Ramp filter, no error.

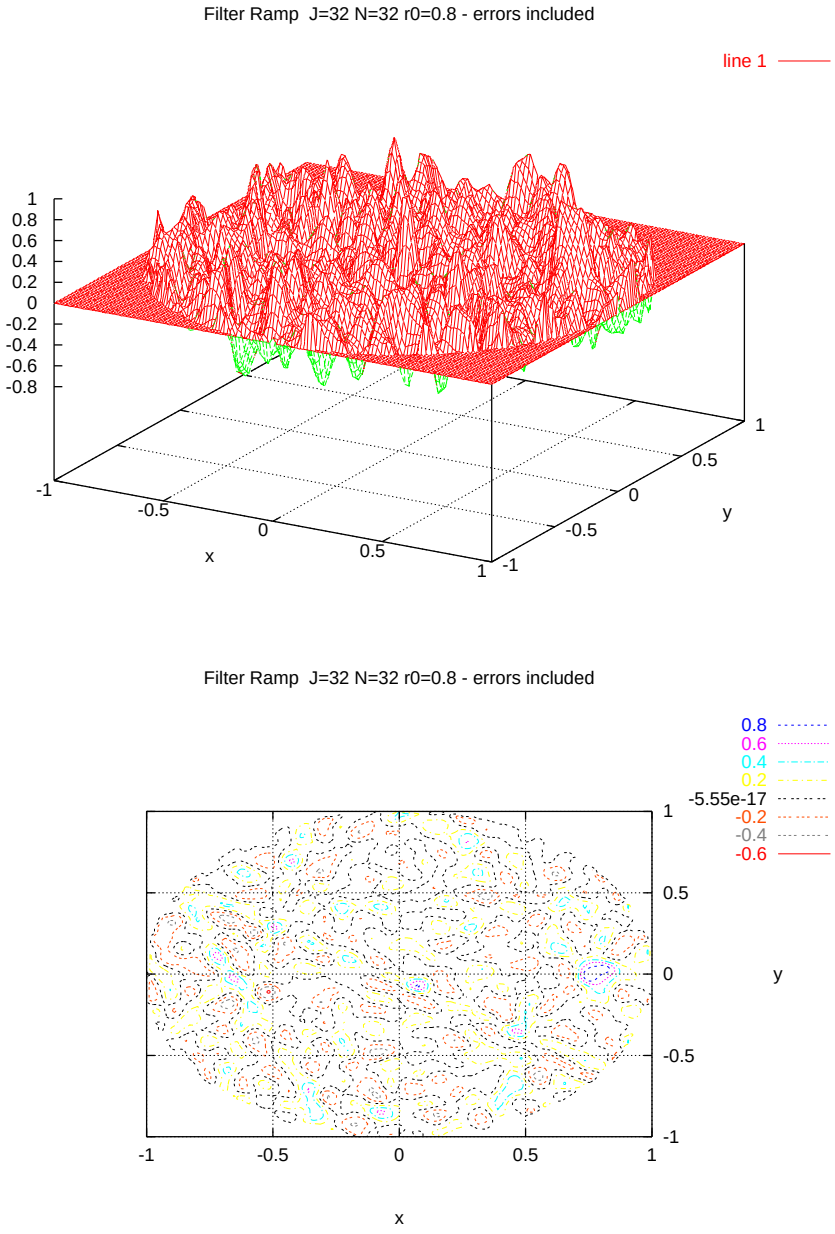


Figure 5.9: Contour and mesh plots, $N = 32, J = 32, r_0 = 0.8$, error added.

In the angular plot figure 5.10, the general shape of the reconstructed point source is again similar to that of figure 4.4. In agreement with the earlier plots, the width of the primary peak depends mainly on J, and is found a factor of 2 larger than that of figure 4.4. The shape of higher order peaks depends as seen earlier mainly on N. This effect is not expessed in figure 4.4. Adding errors yields figure 5.11, but noise influences to such degree, that any conclusions are omitted.
Again, results are similar using the other filters, as documented by appendix B.

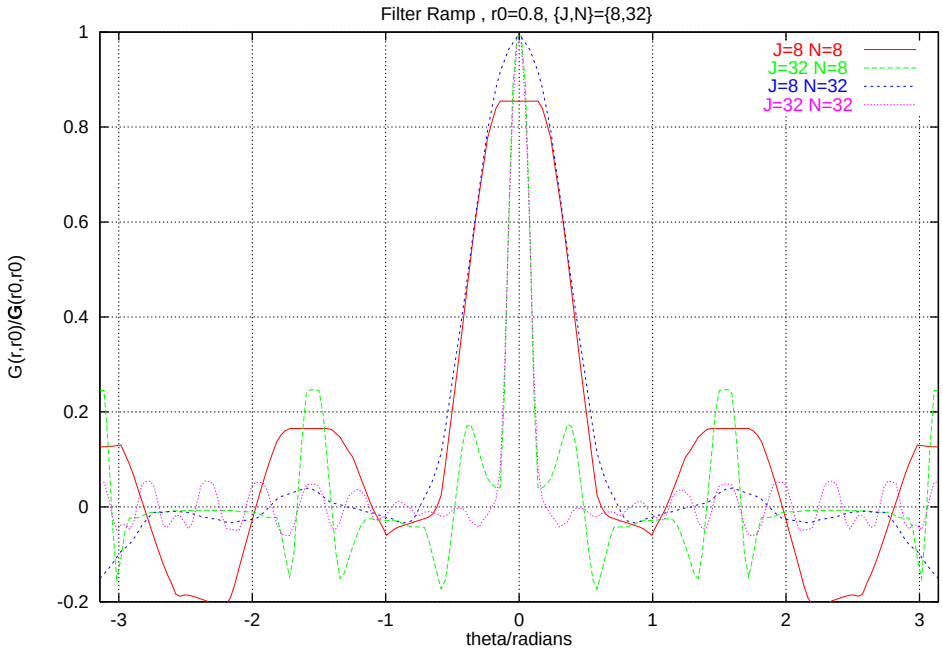


Figure 5.10: $G(r, r_0)$ as fct. of θ, point source positioned at $r_0 = 0.8$.

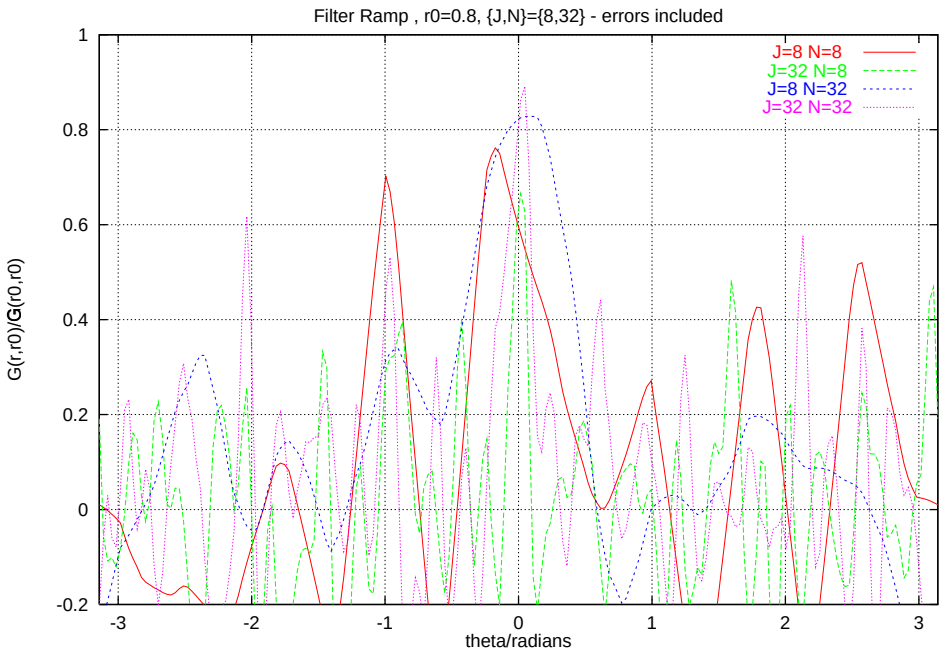


Figure 5.11: As figure 5.10, but errors added.

Using reconstructions simliar to those presented in figures 5.2 – 5.6 several other values of r_0, the dependence of radial width W_r on the point source position r_0 was investigated. Results are shown in figure 5.12. It is seen that the radial width is to good approximation *constant* as function of r_0, in contrast to the result of [15], depicted in figure 4.6. Also note that, in agreement with the previously shown plots, the width is at all values of r_0 found to be a factor of 2 larger than the maximum width seen in figure 4.5. In agreement with the previously shown plots, the width depends mainly on J. For $J = 8$, the width is so large, that the point-spread function reaches beyond the reconsrtuction area. This makes determination of the FWHM width difficult at both ends of the r_0 axis. Results found for the other filters are consistent with these conclusions, as documented by appendix B. Comparing the widths determined for each filter, it is seen that these depend on the used filter. The smallest seen widths are those of the Ramp filter, next the sinc filter, the Hamming filter and finally the cos filter.

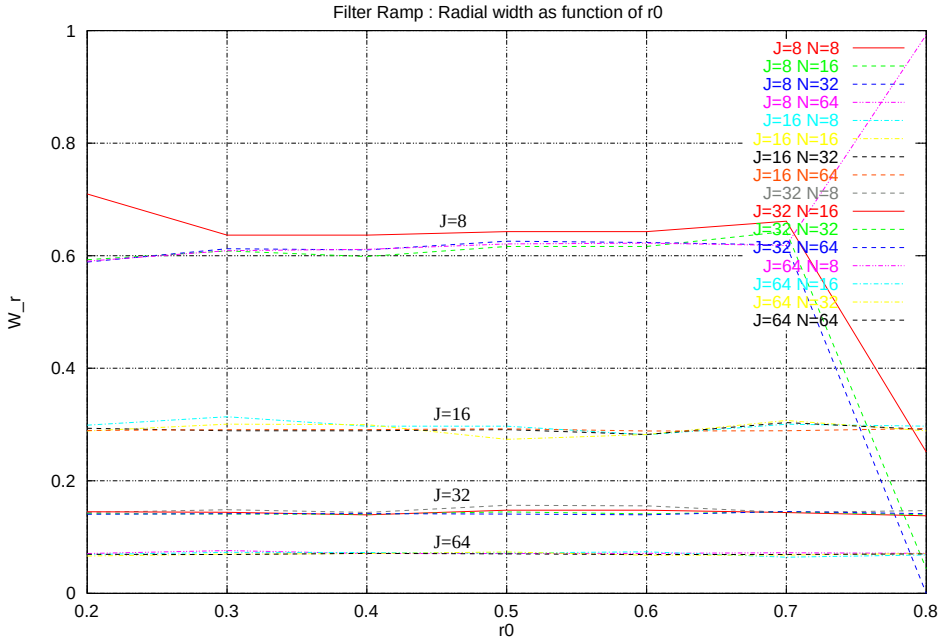


Figure 5.12: Radial width W_r as fct. of r_0.

From a set of images similar to figures 5.10, for point source postitions r_0 ranging from 0.2–0.8, the angular FWHM W_θ was determined as function of r_0. The resulting graphs are shown in figure 5.13. From these data, the corresponding transverse widths W_t were calculated using the relation $W_\theta = 2 \arcsin \frac{W_t}{2r_0}$, as seen in figure 5.14. Note that W_t is constant as function of r_0, and that to good approximation $W_t = W_r$. This result differs from that of [15], presented in figure 4.6. Note that for $J = 8$, the angular width is so large that a valid determination of the angular width is not possible at low values of r_0. Results found for the other filters are consistent with these conclusions, as documented by appendix B. Comparing the widths determined for each filter, it is seen that these depend on the used filter. The smallest seen widths are those of the Ramp filter, next the sinc filter, the Hamming filter and finally the cos filter.

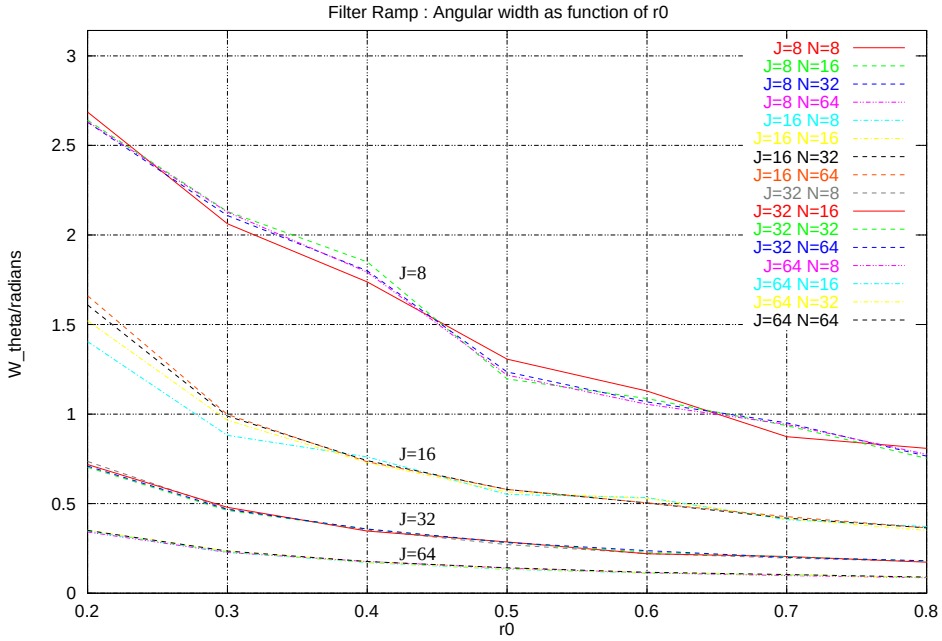


Figure 5.13: Angular width W_θ as fct. of r_0.

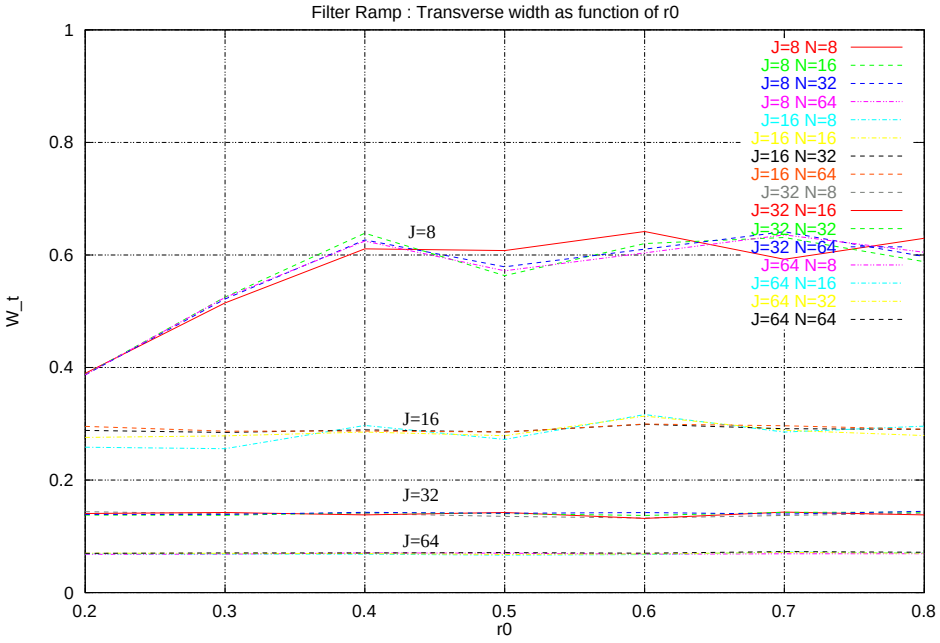


Figure 5.14: Transverse width W_t as fct. of r_0.

5.2.2 Importance of J vs. N

From the figures 5.15–5.16 below, it is easily seen that if the number of angular views N obey $N \approx J$, doubling the value of N makes no significant change to the point-spread function. This is also to some extent documented by the previous figures. In other words, it is not feasible to take a larger number of projections than the number of detectors. The shown result is valid for all of the applied filters, as documented by the datafiles on the enclosed cdrom. For instructions on the use of the cdrom, see appendix C.

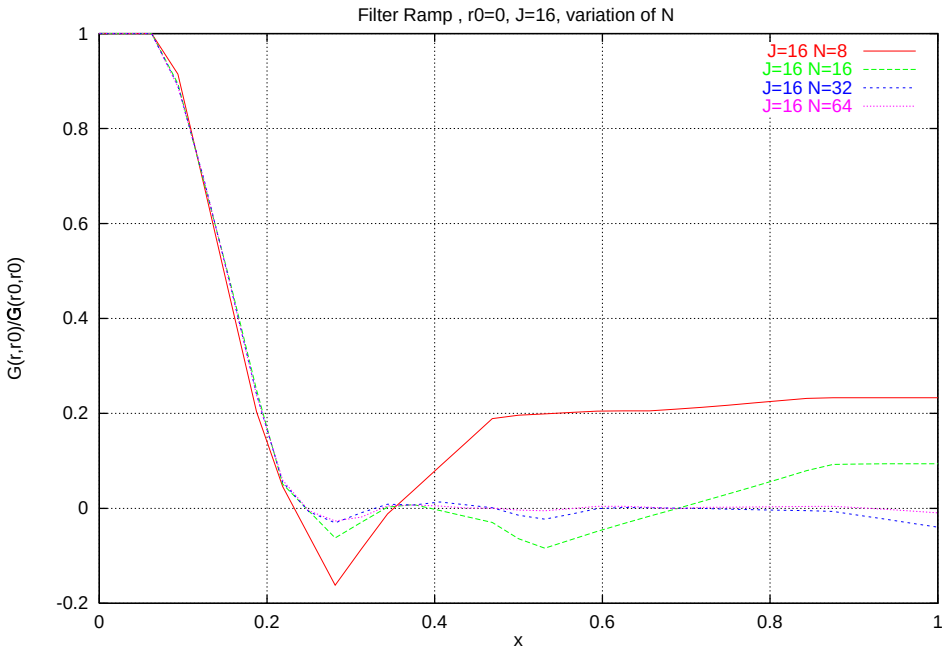


Figure 5.15: Variation of N with $J = 16$ at $r_0 = 0$.

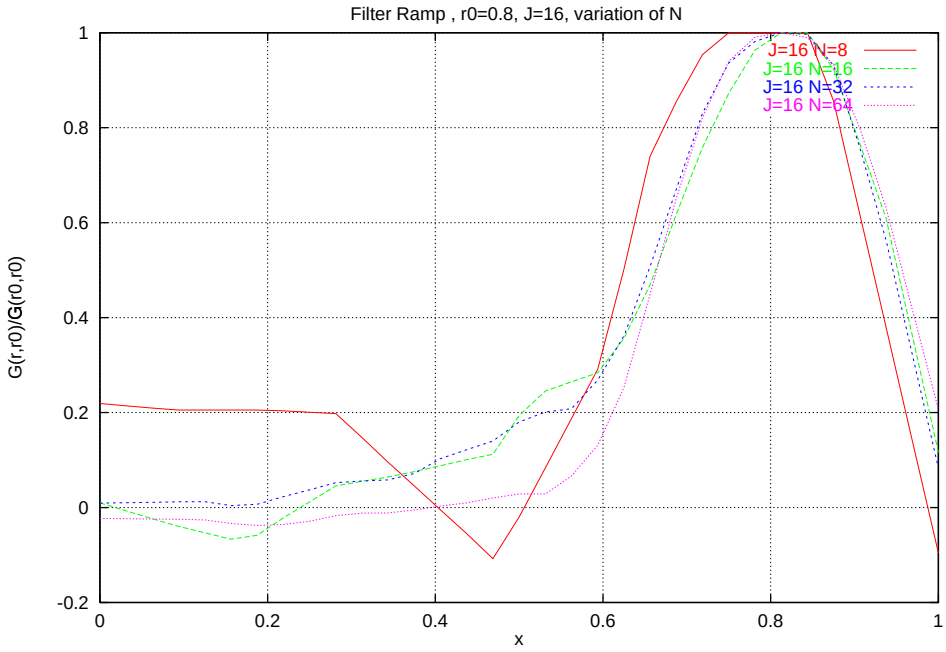


Figure 5.16: Variation of N with $J = 16$ at $r_0 = 0.8$.

5.2.3 Result summary

Summarising, we have found that the width of the point-spread function depends mainly on J - the number of detectors, which is very different from the results of [15]. The found widths are a factor of 2 larger than the maximum widths found in [15]. The number of angular views N influences secondary effects, and this influence is partly removed when errors are included. The shape of the point-spread function is found have common features with that of [15]. The conclusions stated here apply to all of the investigated filters, as documented by appendix B. The width of the point-spread function depends on the used filter, the width is seen smallest for the Ramp filter, next the sinc filter, the Hamming filter and finally largest for the cos filter. Interference effects or higher order peaks depend on the used filter, in agreement with section 2.3.1. It has been demonstrated that when building a scanner of the geometry shown in figure 2.2, the important parameter is J in favour of N. It is not feasible to have a scanner with the number of angular views N much higher than the number of detectors J.

Since the studies on the effect of noise on reconstruction were not carried out systematically, this could be a subject of further investigation. Also, the influence of scanner geometry different from the one assumed in this thesis (figure 2.2) should be investigated further. These topics could perhaps serve as the foundation of another Cand. Scient. thesis, enhancing and developing the results of this thesis.

Chapter 6

Conclusion

A thorough description of the reconstruction mathematics involved in Filtered Back-Projection (FBP) has been given. The presented theory, the analytical reconstruction examples and the chapter on scanner technology together form a broad overview of the field of medical tomographical imaging.

On the results of the simulation study, the following conclusions are made. As shown in section 5.2, the general shape of the reconstructed point source is found to be similar to that of [15]. The primary parameter characterising the point-spread function, namely the width of the reconstructed source, is generally a factor of 2 larger in the simulation study than in the theory of [15], found in section 4.1. The width is found to depend on the number of detectors J only, in contrast to [15] where the width depends on the point source position r_0. Widths are seen to depend on the used filter, smallest width seen for the Ramp filter, next the sinc filter, the Hamming filter and finally largest for the cos filter. Higher order peaks are found to mainly depend on the number of angular views N, a dependence which is partly removed when errors are added. The used filter also influence the shape of higher order peaks, in agreement with section 2.3.1. The transverse width W_t is found equal to the radial width W_r, which is not seen in [15]. It has also been shown that it is not feasible to increase the number of angular views N to a value much higher than the number of detectors J.

The paper [15] states to have found the *best possible* or narrowest, general point-spread function for tomographic imaging. Since the widths found in the simulation study are a factor of 2 *larger* than the width of the point-spread function of [15], the used algorithm of the simulation study *could* be an example of a reconstruction method *less accurate* than the point-spread

function of [15], which could be argued to support the findings of [15]. On the other hand, the found dependence of r_0 on the width in the simulation study differs from the dependence found in [15]. This difference talks against the validity of [15]. Hence, the results of the simulation study are not clear enough to reject nor confirm the findings of [15].

With these final remarks I conclude my thesis for the Cand. Scient. degree.

Frederiksberg, February 25th 2000.

Peter Willendrup

Appendix A

Zernike Polynomials

The polynomials were introduced by Zernike in 1934 and some of their properties are given by Born and Wolf ([24]). Notation and properties follow.

Orthogonality

$$\begin{aligned} \int_0^1 r\,dr R_n^l(r) R_n^\lambda(r) &= h_n^l \delta_{l\lambda} \qquad \text{(A.1)} \\ h_n^l &= \frac{1}{2(n+2l+1)} \end{aligned}$$

Recursion formulae

$$a_n^l R_n^{l+1}(r) = (b_n^l - c_n^l r^2) R_n^l(r) - d_n^l R_n^{l-1} \qquad \text{(A.2)}$$

$$\begin{aligned} a_n^l &= (n+l+1)(n+2l)(l+1) \\ b_n^l &= -(n+2l+1)[(n+l)(n+l+1)+l(l+1)] \\ c_n^l &= -(n+2l)(n+2l+1)(n+2l+2) \\ d_n^l &= (n+2l+2)(n+l)l \end{aligned}$$

$$(n+2l) r R_{n-1}^l(r) = (n+l) R_n^l(r) + l R_n^{l-1}(r) \qquad \text{(A.3)}$$

Explicit expression

$$R_n^l(r) = r^n \sum_{s=0}^{l} (-1)^{l-s} \frac{(n+l+s)!}{(l-s)!s!(n+s)!} r^{2s} \qquad \text{(A.4)}$$

Special values

$$\begin{array}{lcllcl} R_n^l(0) & = & (-1)^l\delta_{n0} & R_n^0(r) & = & r^n \\ R_n^l(1) & = & 1 & R_n^1(r) & = & r^n((n+2)r^2-(n+1)) \end{array}$$

Cristoffel-Darboux formula

$$\sum_{l=0}^{J-1}\frac{1}{h_n^l}R_n^l(r)R_n^l(r_0) = 2\frac{J(J+n)}{2J+n}\,\frac{R_n^J(r_0)R_n^{J-1}(r)-R_n^J(r)R_n^{J-1}(r_0)}{r_0^2-r^2} \tag{A.5}$$

Roderique's formula

$$R_n^l(r) = \frac{1}{l!r^n}\left(\frac{d}{d(r^2)}\right)^l\{(r^2)^{l+n}(r^2-1)^l\} \tag{A.6}$$

Relation to other polynomials

$$R_n^l = \left(\begin{array}{c}(n+2l)\\ l\end{array}\right) r^n\, G_l(n+1,n+1,r^2) \tag{A.7}$$

where $G_l(p,q,x)$ is a Jacoby polynomial (reference([25]) chapter 22.)

Generating function

$$\frac{(1+z-\sqrt{1+2z(1-2r^2)+z^2})^n}{(2zr)^n\sqrt{1+2z(1-2r^2)+z^2}} = \sum_{l=0}^{\infty} z^l\, R_n^l(r) \tag{A.8}$$

Integrals

$$\int_s^1 \frac{rdr}{\sqrt{r^2-s^2}}\, R_n^l(r)T_n(s/r) = \frac{1}{n+2l+1}\sqrt{1-s^2}\, U_{n+2l}(s) \tag{A.9}$$

T_n and U_n are Chebyshev polynomials of the first and second kind respectively.

$$\int_0^r \frac{ds}{\sqrt{r^2-s^2}}\, T_n(s/r)\, U_{n+2l}(s) = \frac{\pi}{2}\, R_n^l(r) \tag{A.10}$$

Note

Notation is different from that of Born and Wolf. What is here called $R_n^l(r)$ corresponds to $R_{n+2l}^n(r)$ in Born and Wolf.

Appendix B

Complementary data

In the three following sections, data obtained with the sinc, Hamming and cos filters are presented. The details on axis units and notation mentioned in section 5.2, also applies to the graphs depicted in this appendix.

B.1 Results obtained with the sinc filter

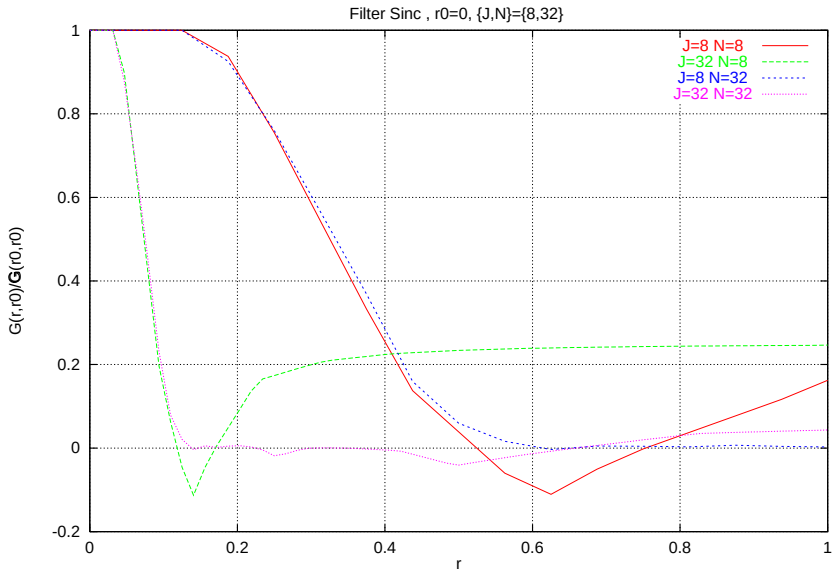


Figure B.1: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0$.

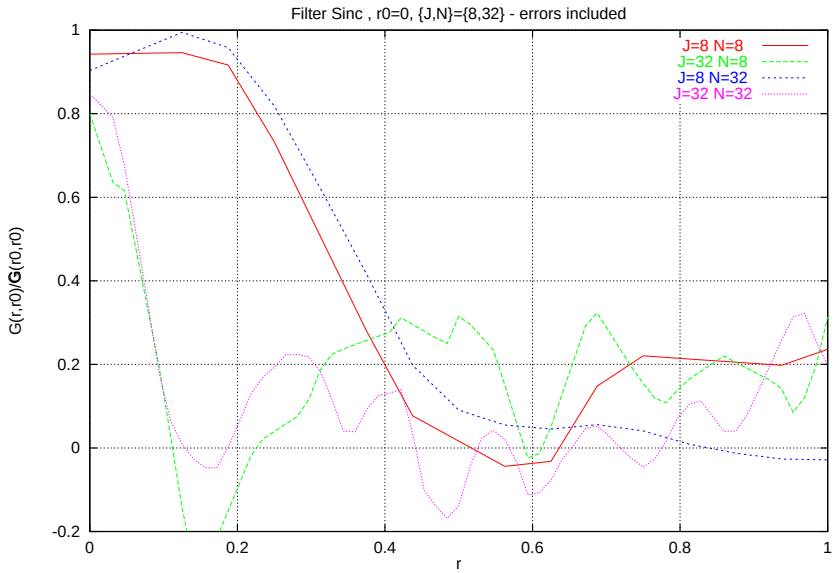


Figure B.2: As figure B.1, but errors added.

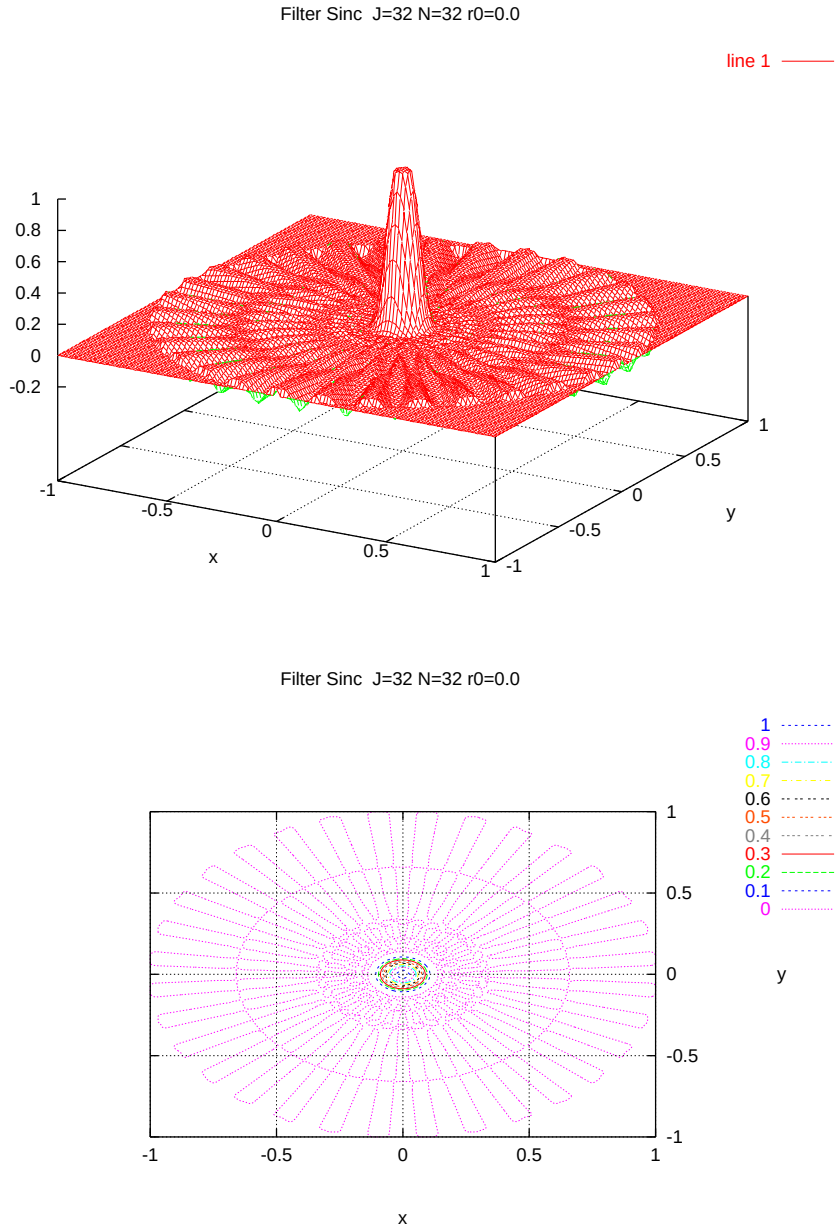


Figure B.3: Contour and mesh plots, N=32, J=32, $r_0 = 0$, no error.

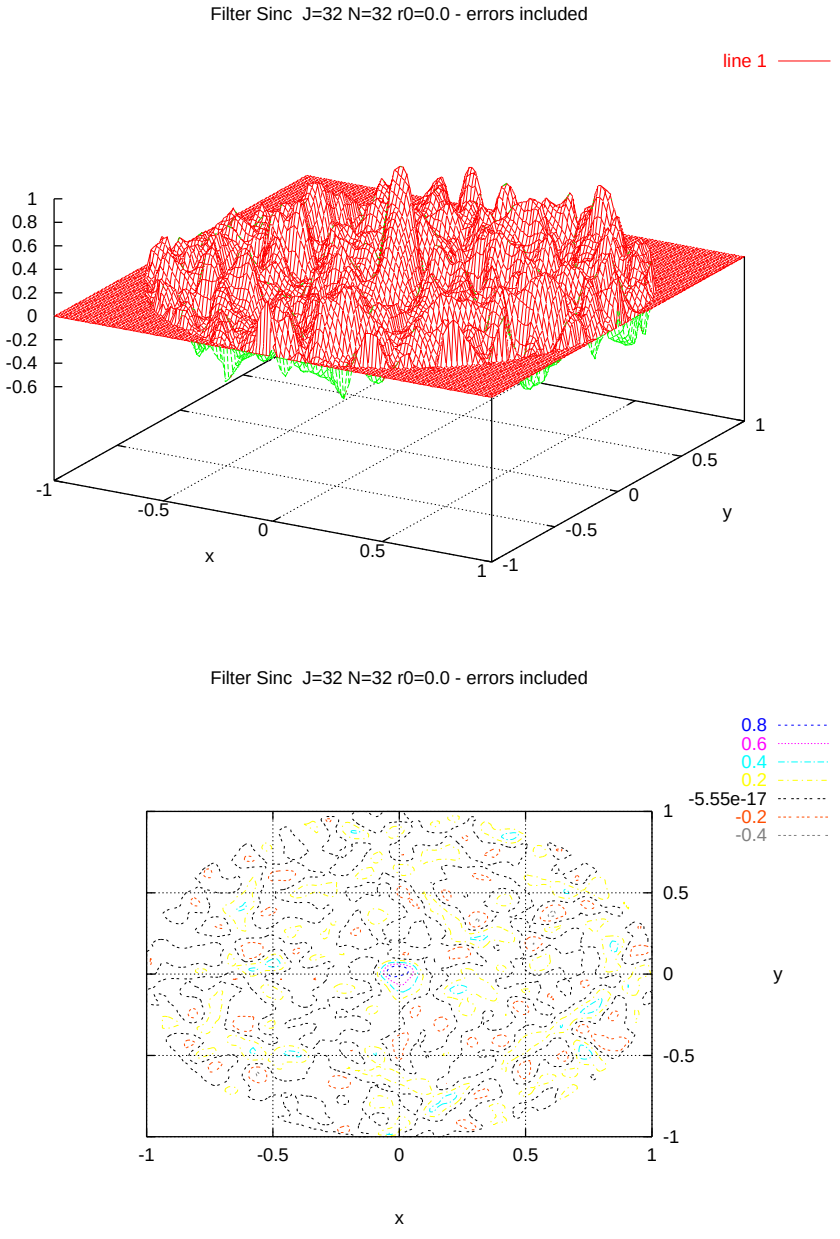


Figure B.4: Contour and mesh plots, N=32, J=32, $r_0 = 0$, error added.

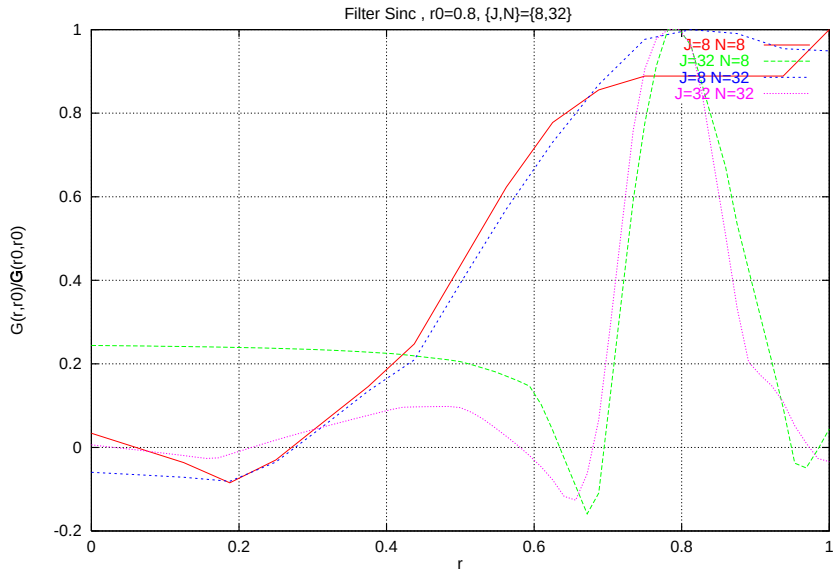


Figure B.5: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0.8$.

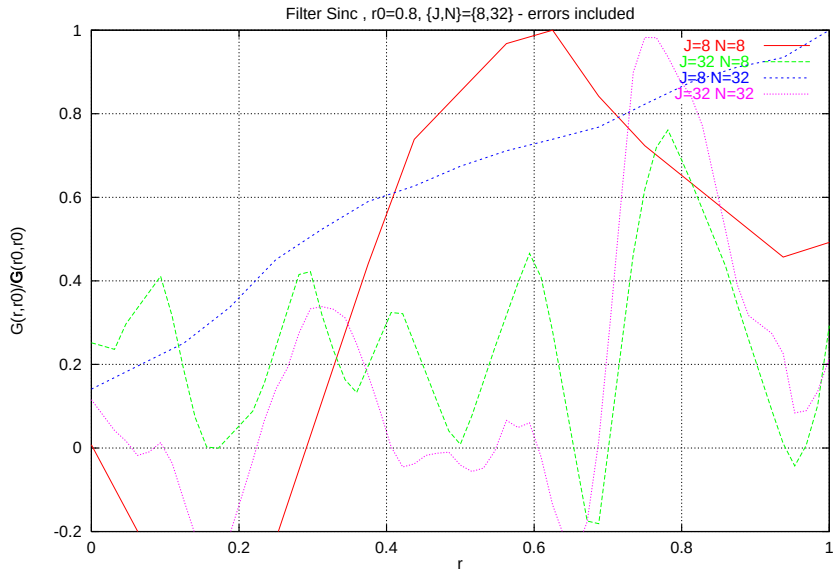


Figure B.6: As figure B.5, but errors added.

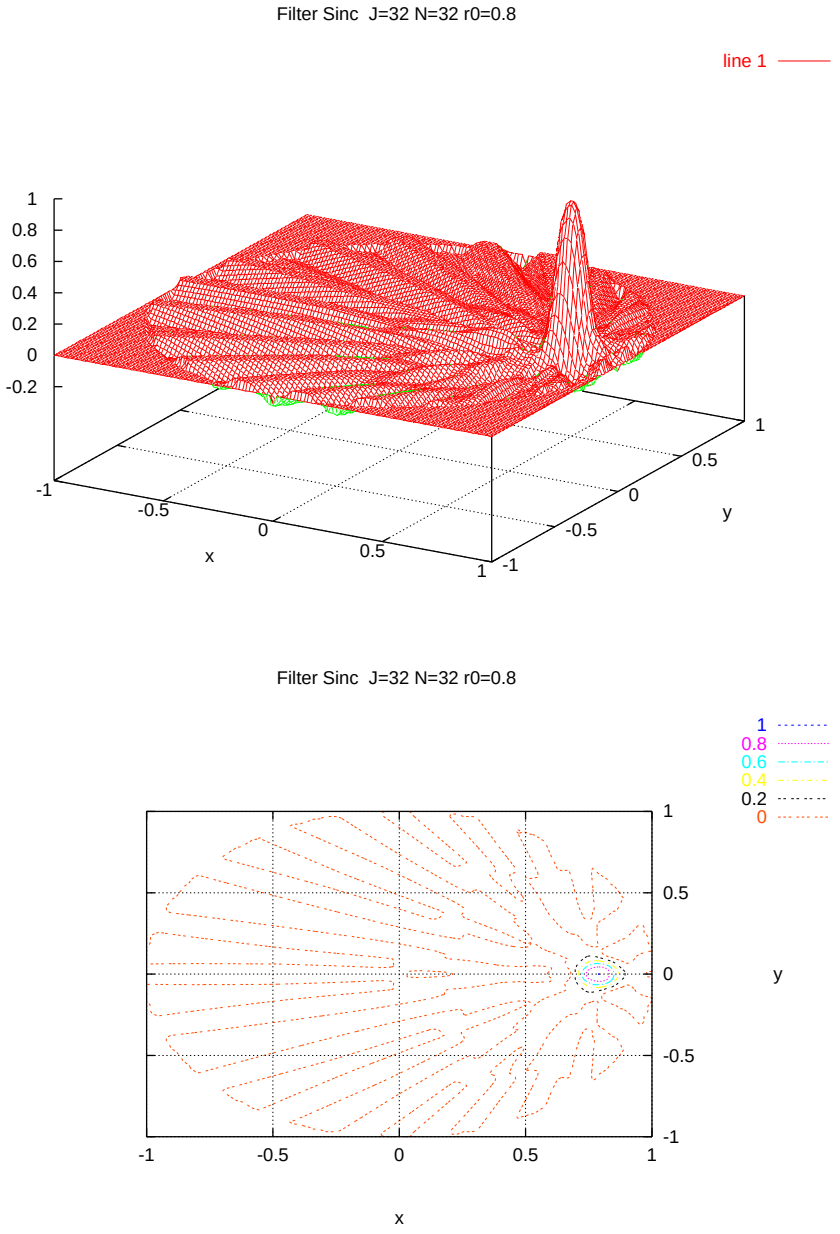


Figure B.7: Contour and mesh plots, N=32, J=32, $r_0 = 0.8$, no error.

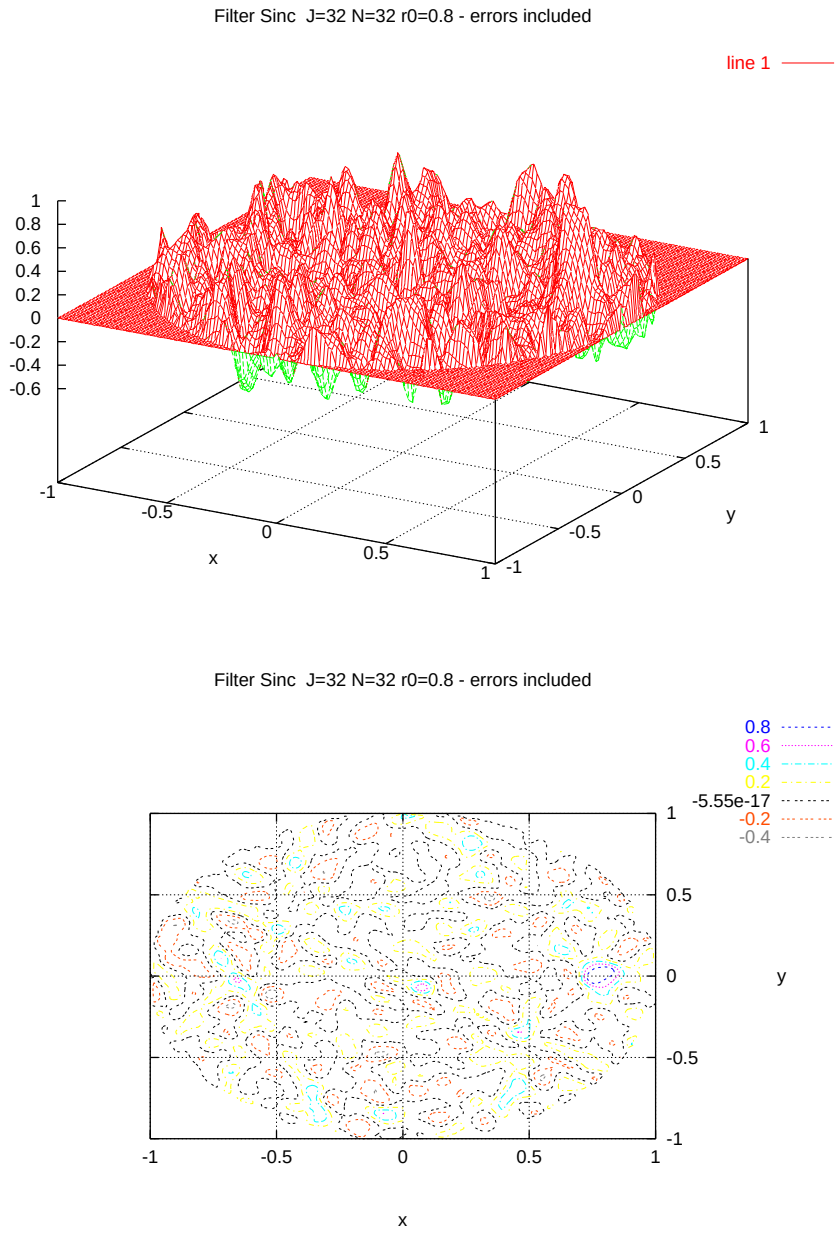


Figure B.8: Contour and mesh plots, N=32, J=32, $r_0 = 0.8$, error added.

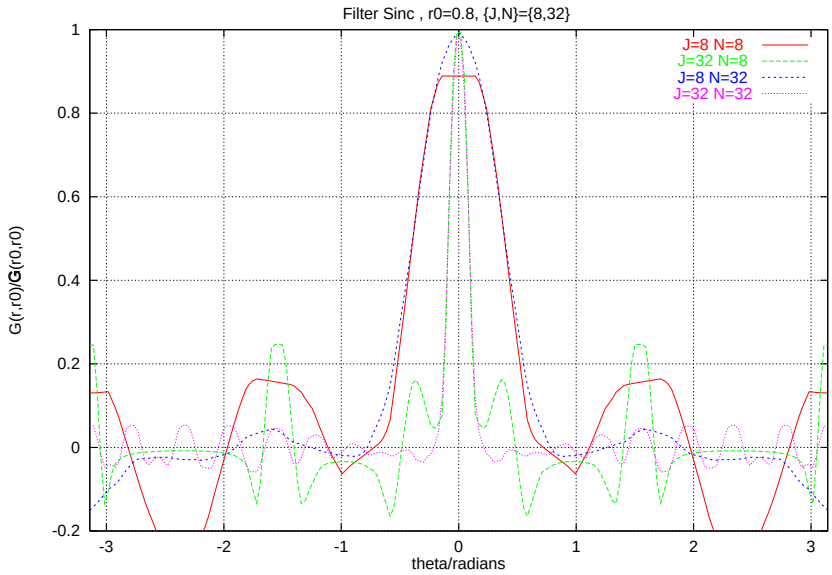


Figure B.9: $G(r, r_0)$ as fct. of θ, point source positioned at $r_0 = 0.8$.

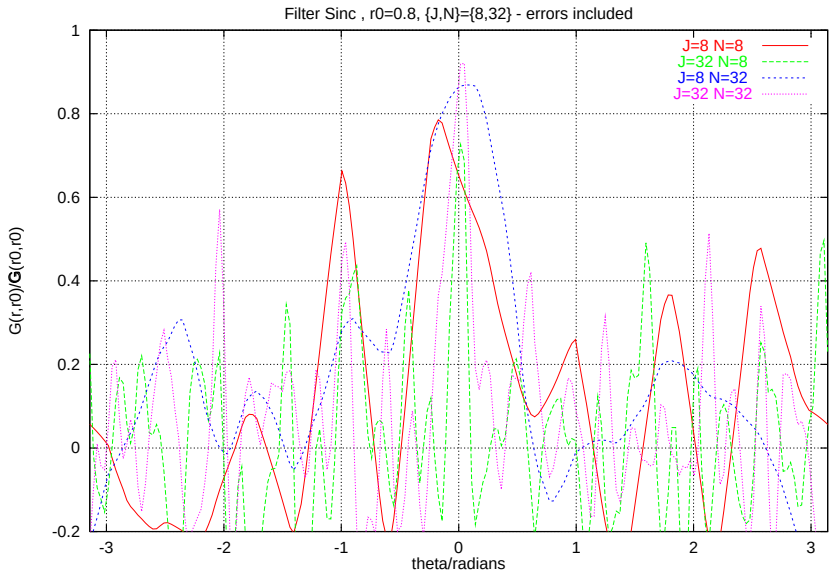


Figure B.10: As figure B.9, but errors added.

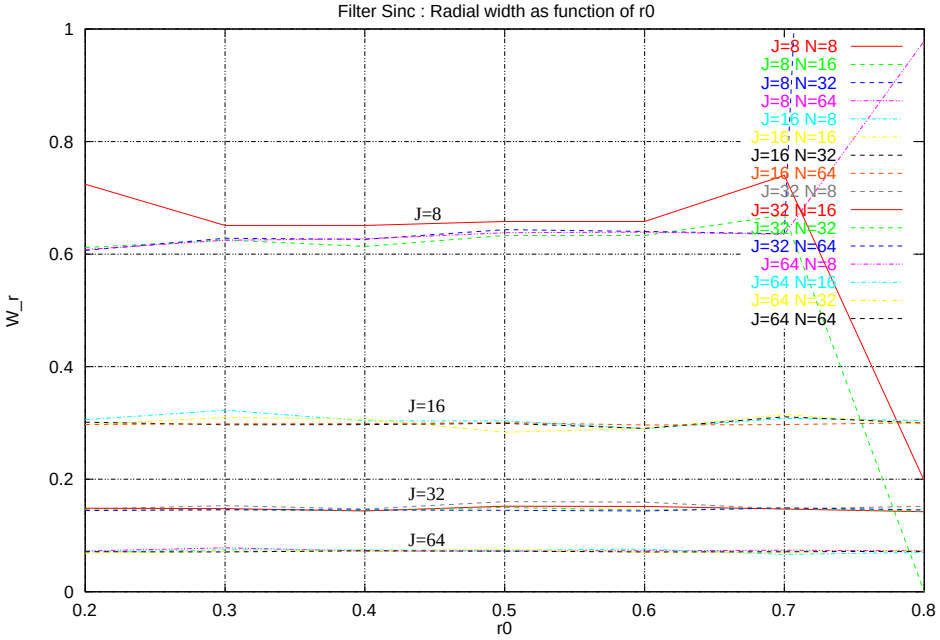


Figure B.11: Filter sinc radial width W_t as fct. of r_0.

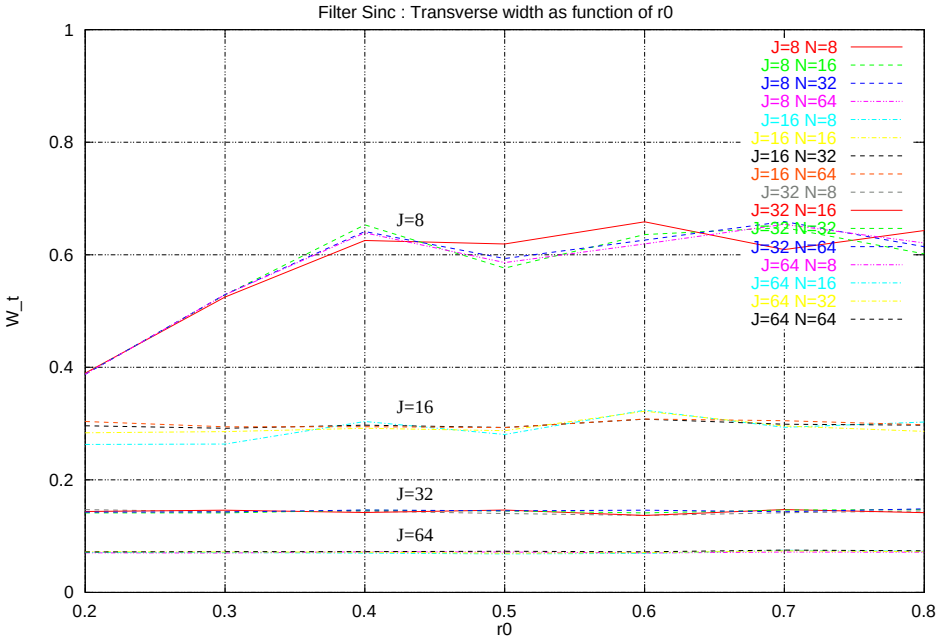


Figure B.12: Filter sinc transverse width W_t as fct. of r_0.

B.2 Results obtained with the Hamming filter

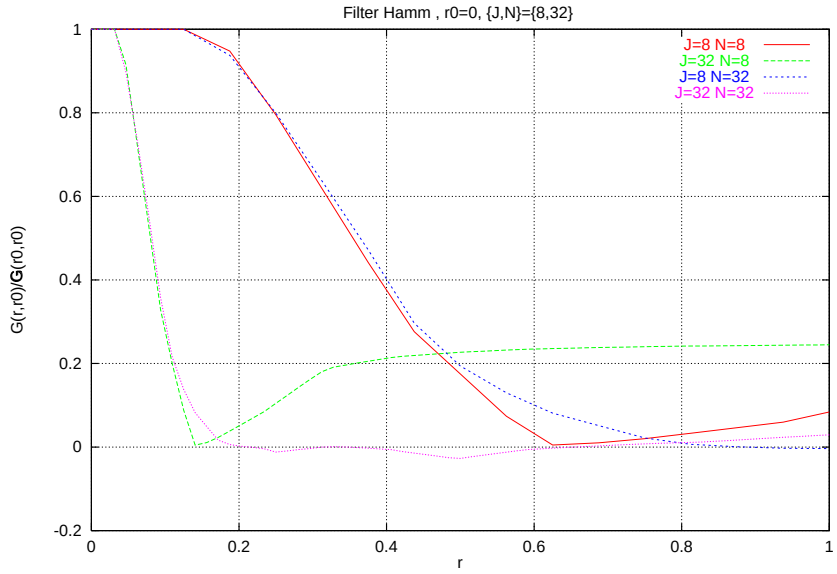


Figure B.13: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0$.

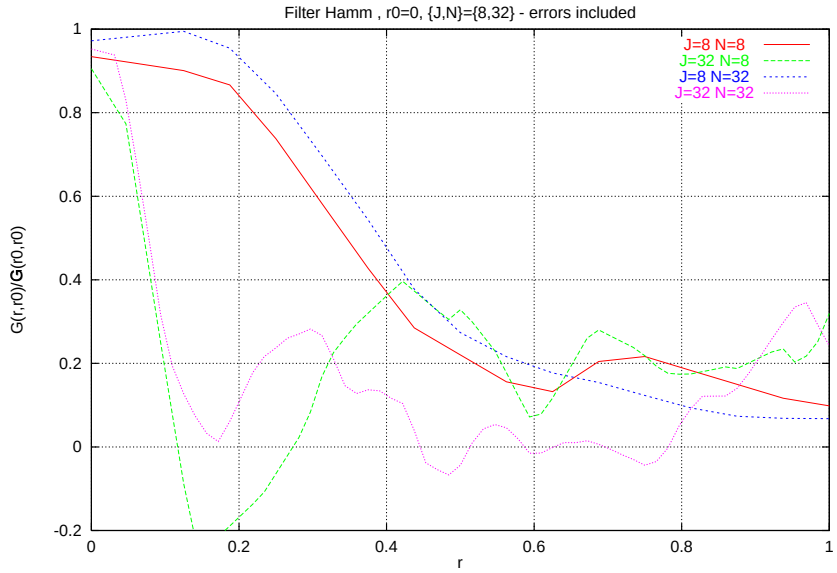


Figure B.14: As figure B.13, but errors added.

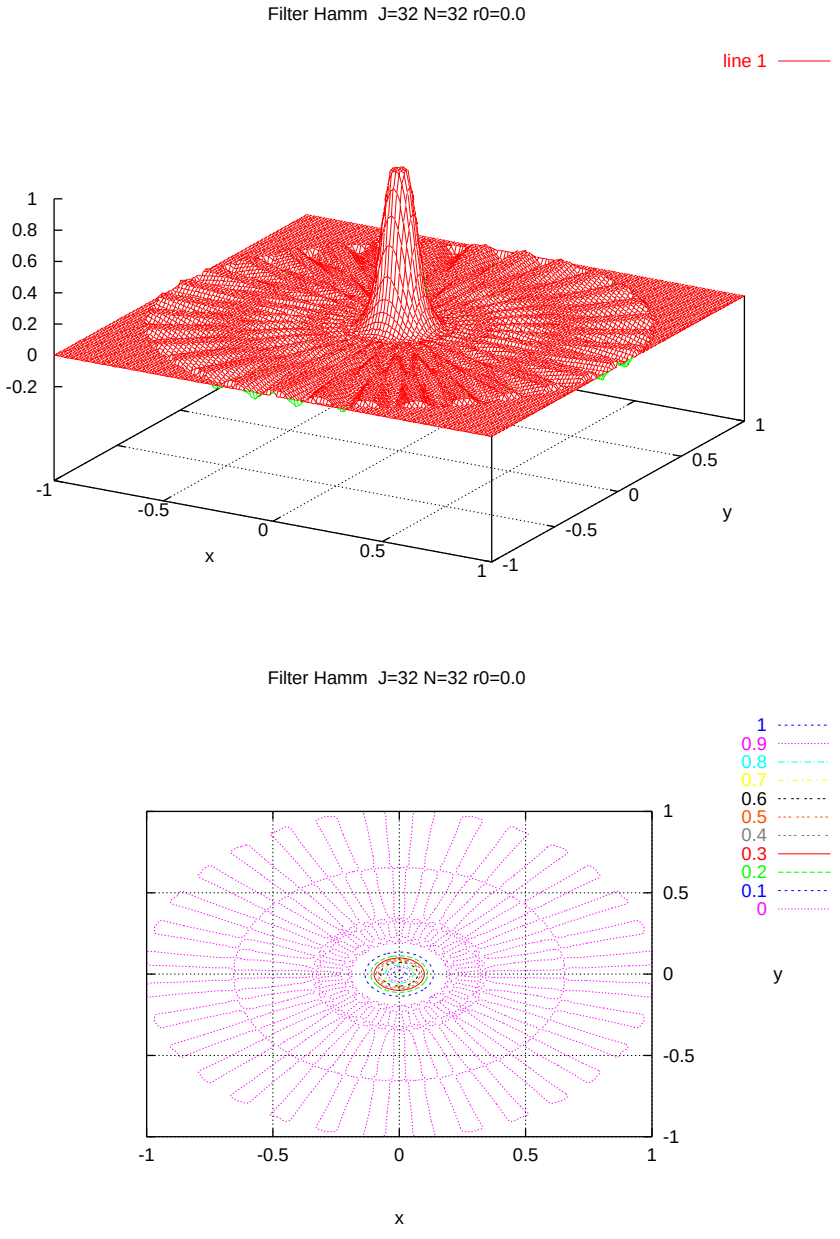


Figure B.15: Contour and mesh plots, $N = 32, J = 32, r_0 = 0$, no error.

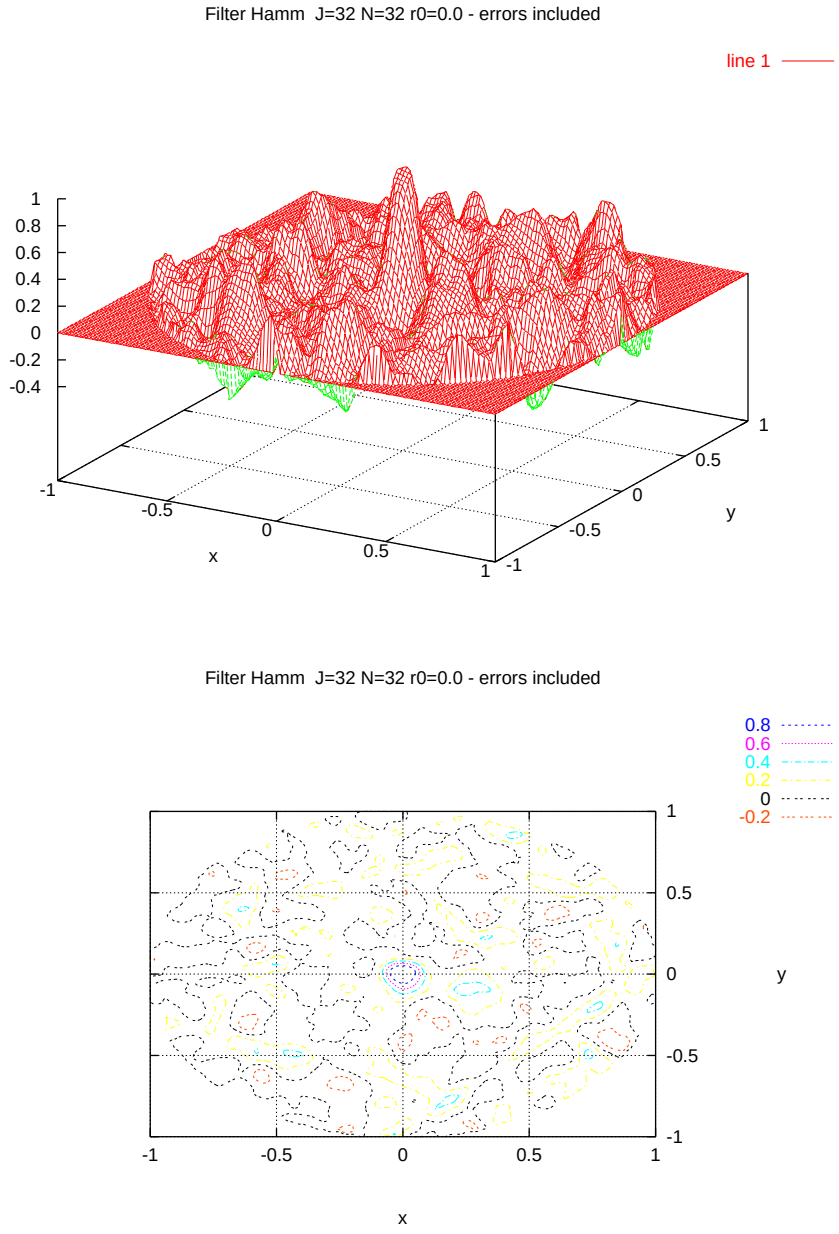


Figure B.16: Contour and mesh plots, $N = 32, J = 32, r_0 = 0$, error added.

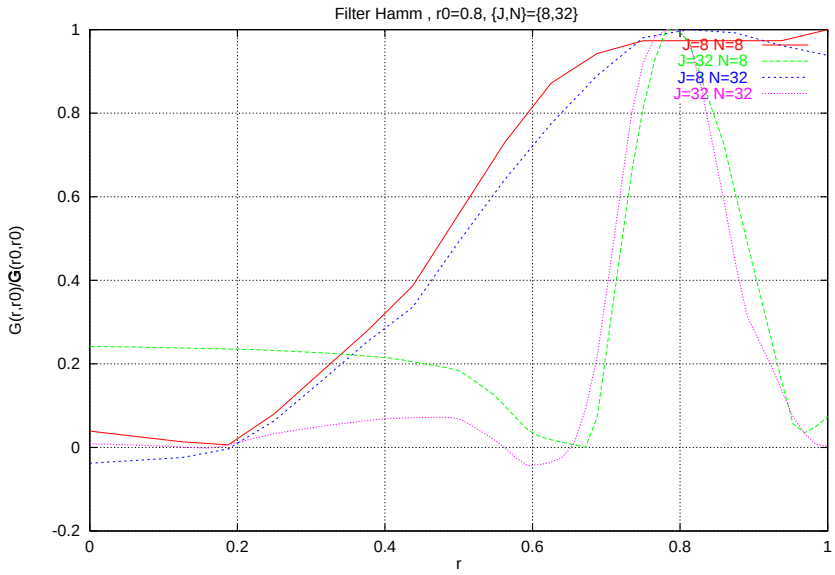


Figure B.17: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0.8$.

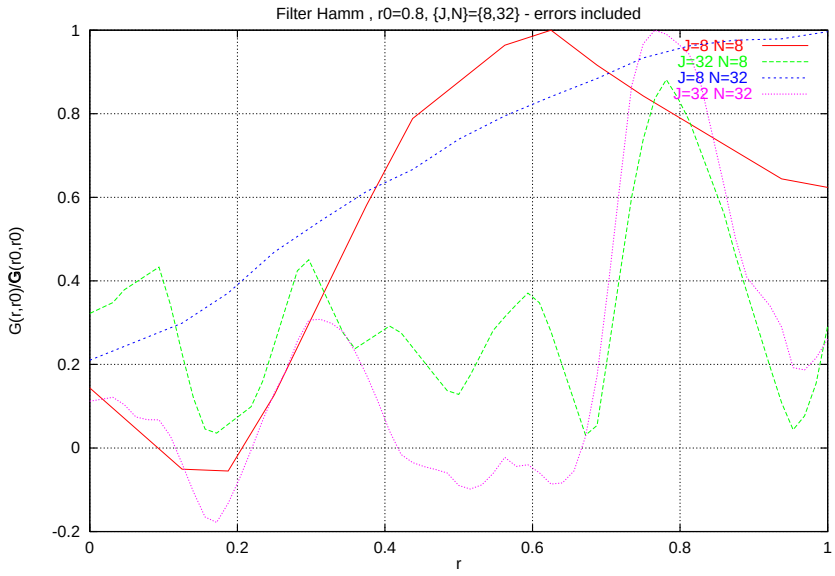


Figure B.18: As figure B.17, but errors added.

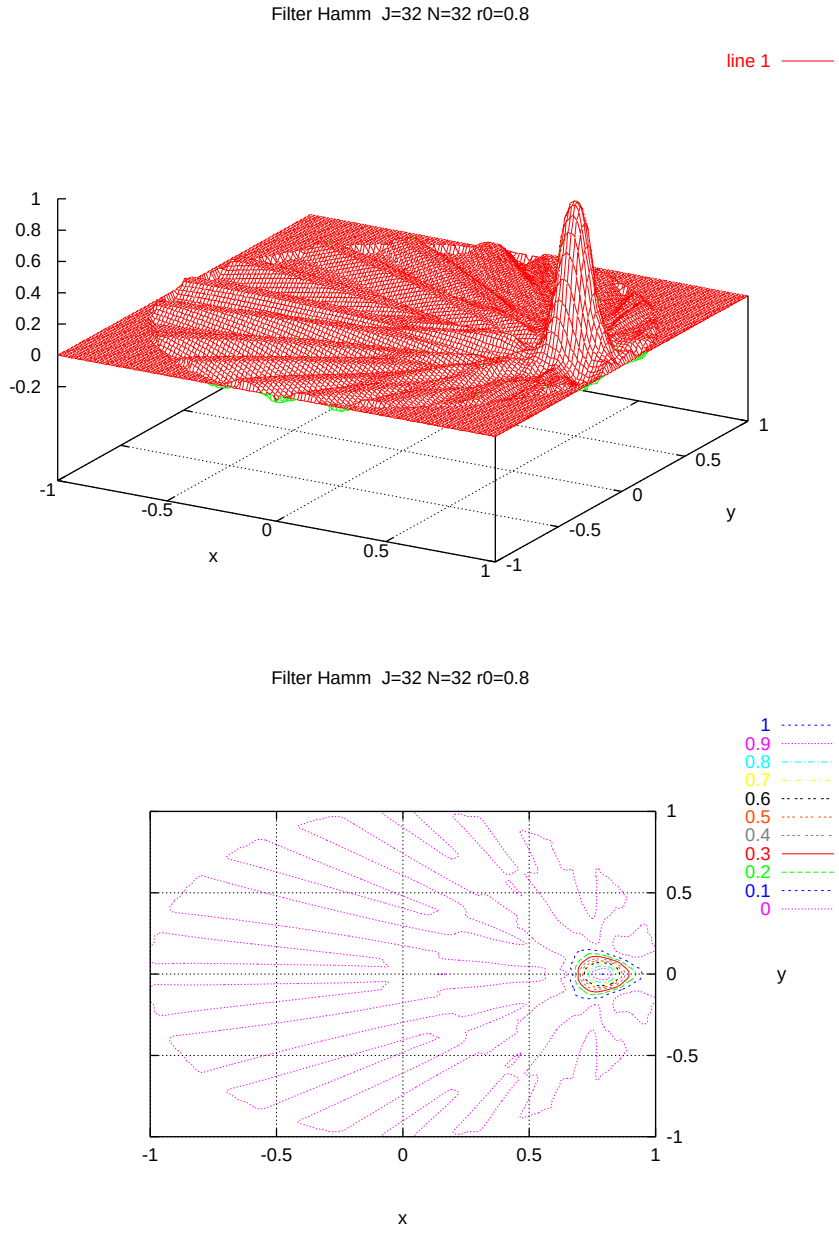


Figure B.19: Contour and mesh plots, $N = 32, J = 32, r_0 = 0.8$, no error.

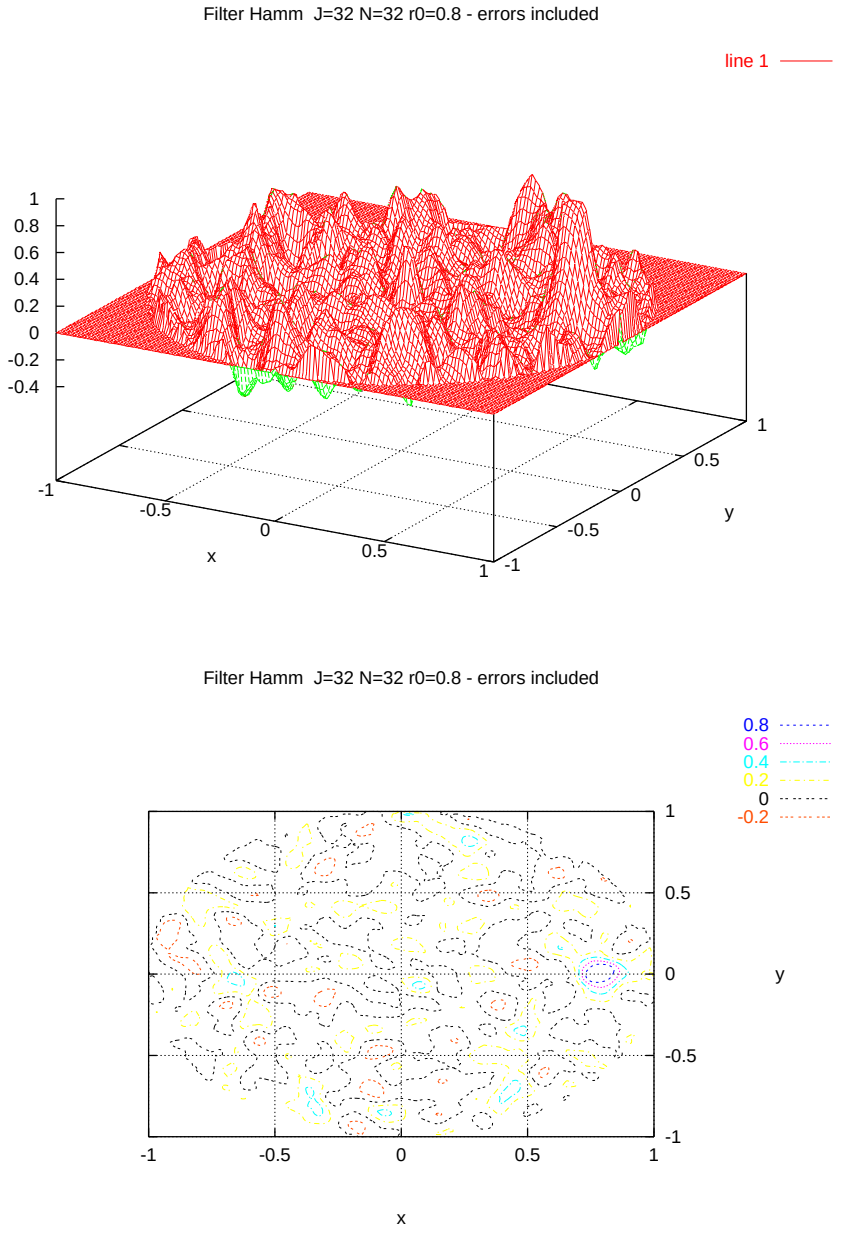


Figure B.20: Contour and mesh plots, $N = 32, J = 32, r_0 = 0.8$, error added.

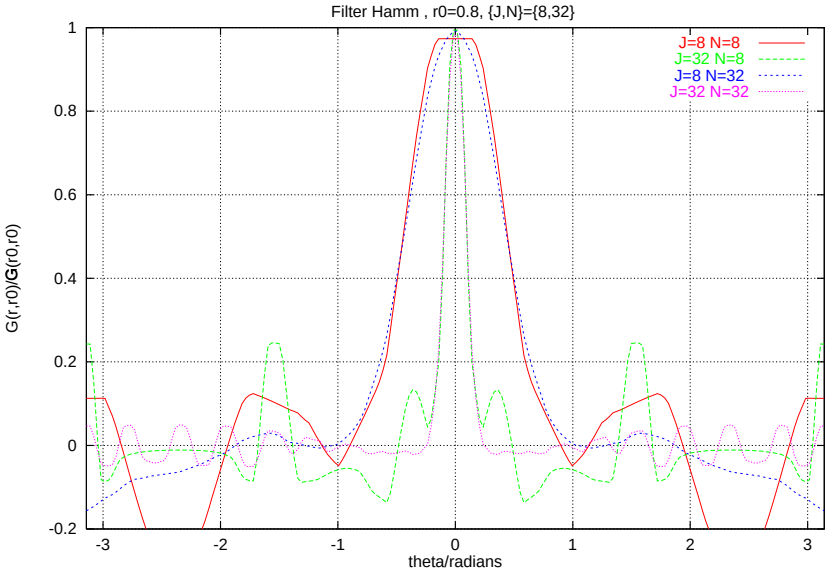


Figure B.21: $G(r, r_0)$ as fct. of θ, point source positioned at $r_0 = 0.8$.

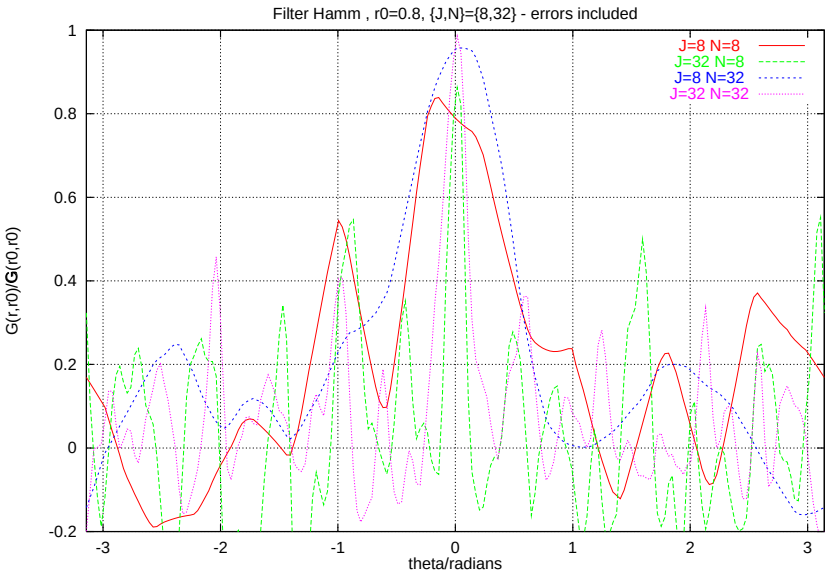


Figure B.22: As figure B.21, but errors added.

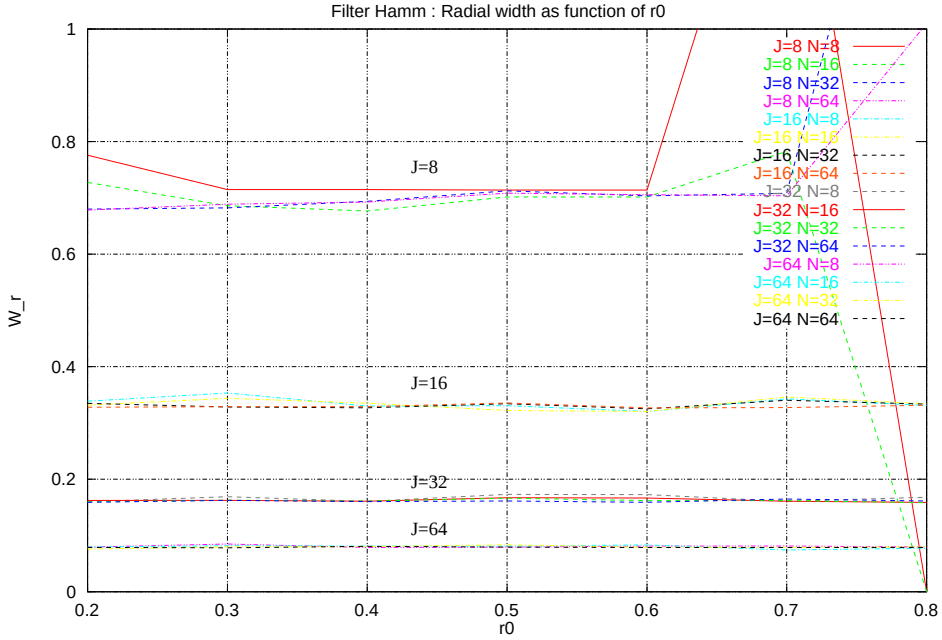


Figure B.23: Radial width W_t as fct. of r_0.

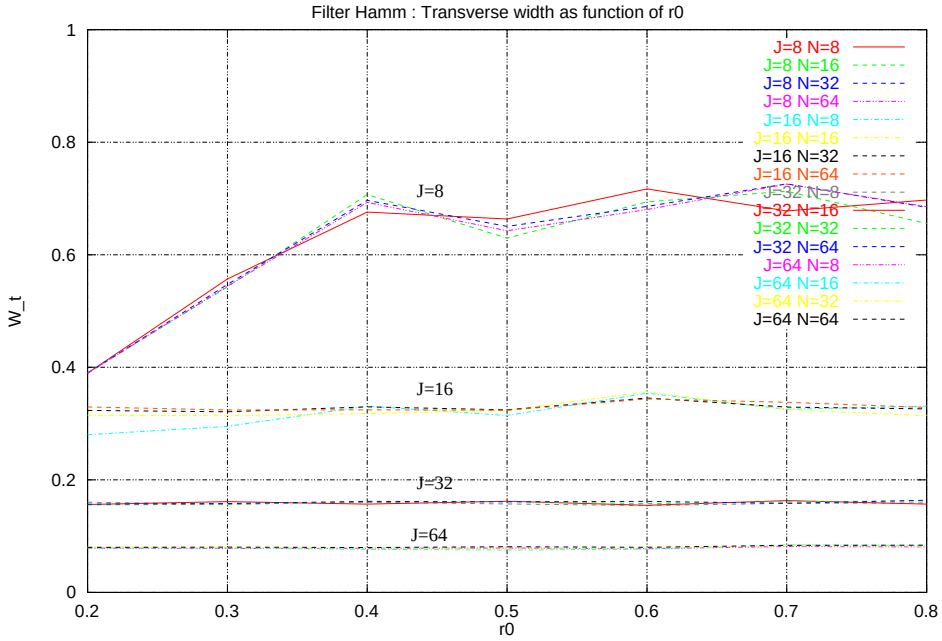


Figure B.24: Transverse width W_t as fct. of r_0.

B.3 Results obtained with the cos filter

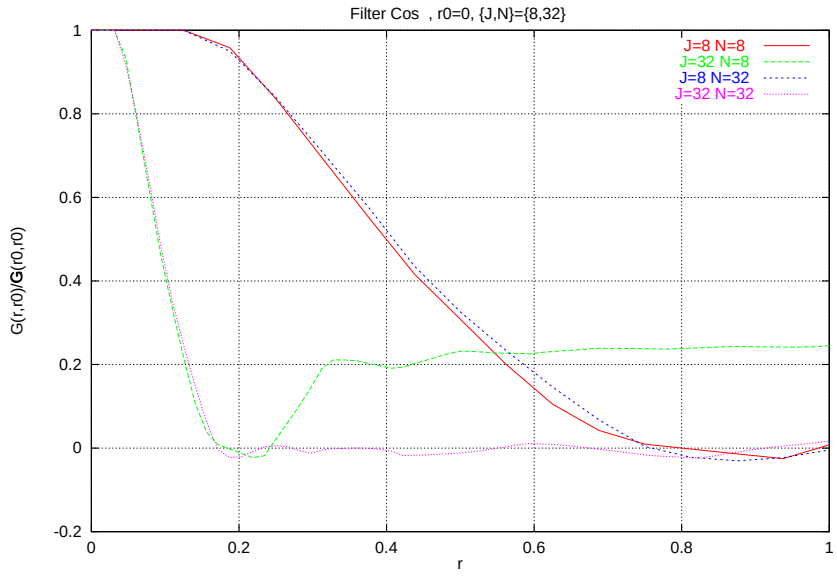


Figure B.25: $G(r, r_0)$as fct. of r, point source positioned at $r_0 = 0$.

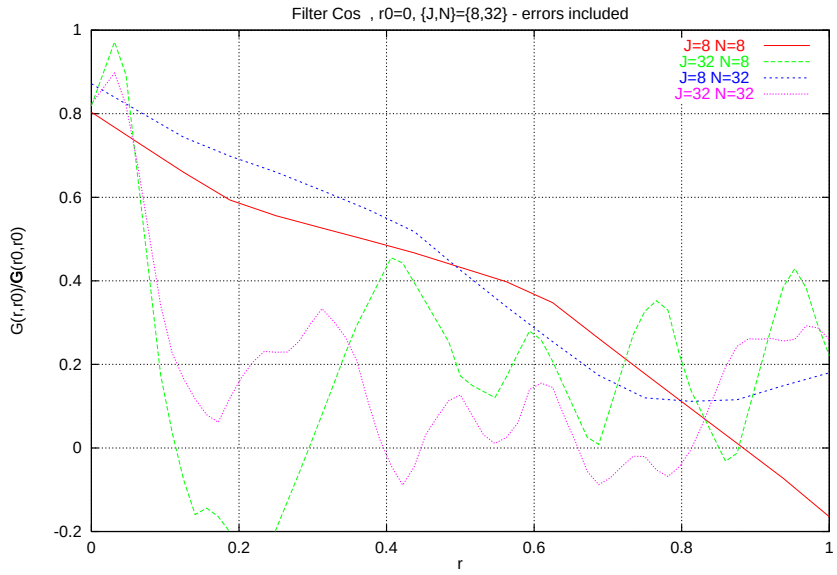


Figure B.26: As figure B.25, but errors added.

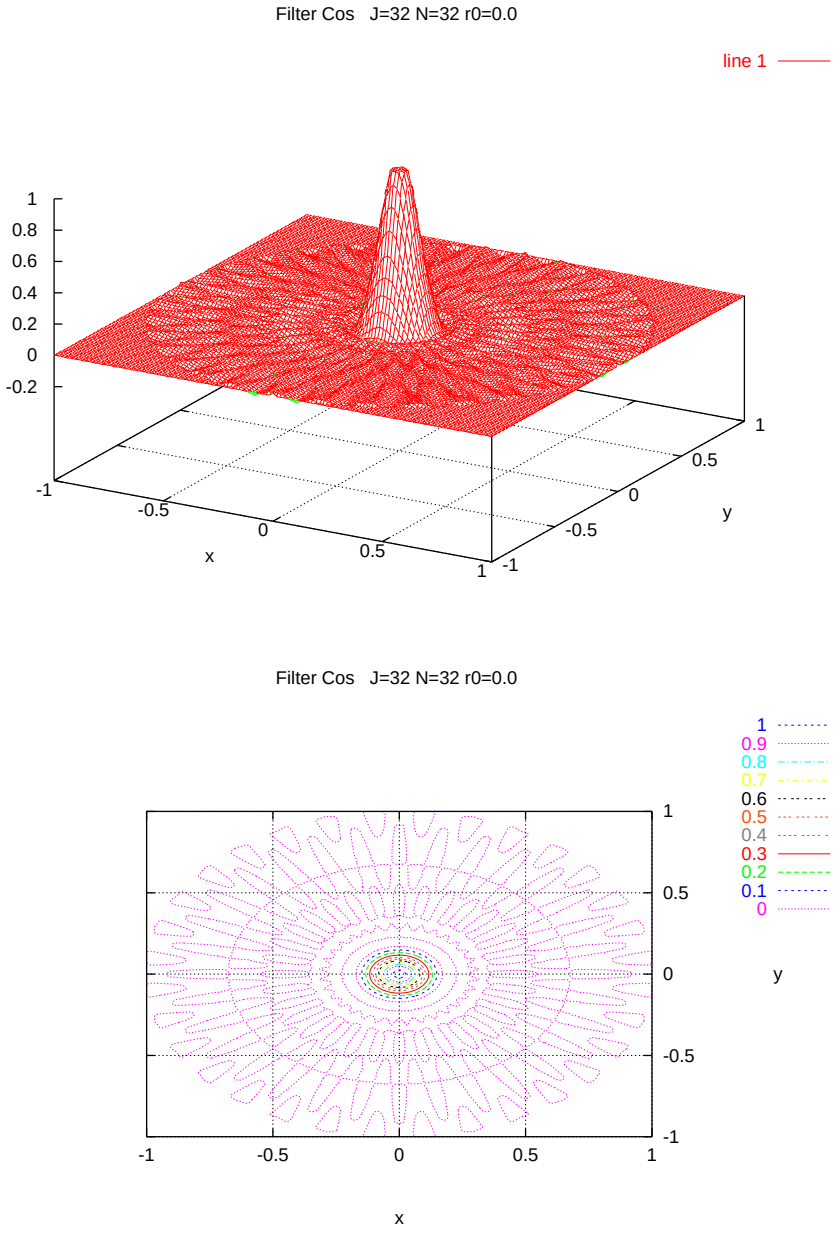


Figure B.27: Contour and mesh plots, $N = 32, J = 32, r_0 = 0$, no error.

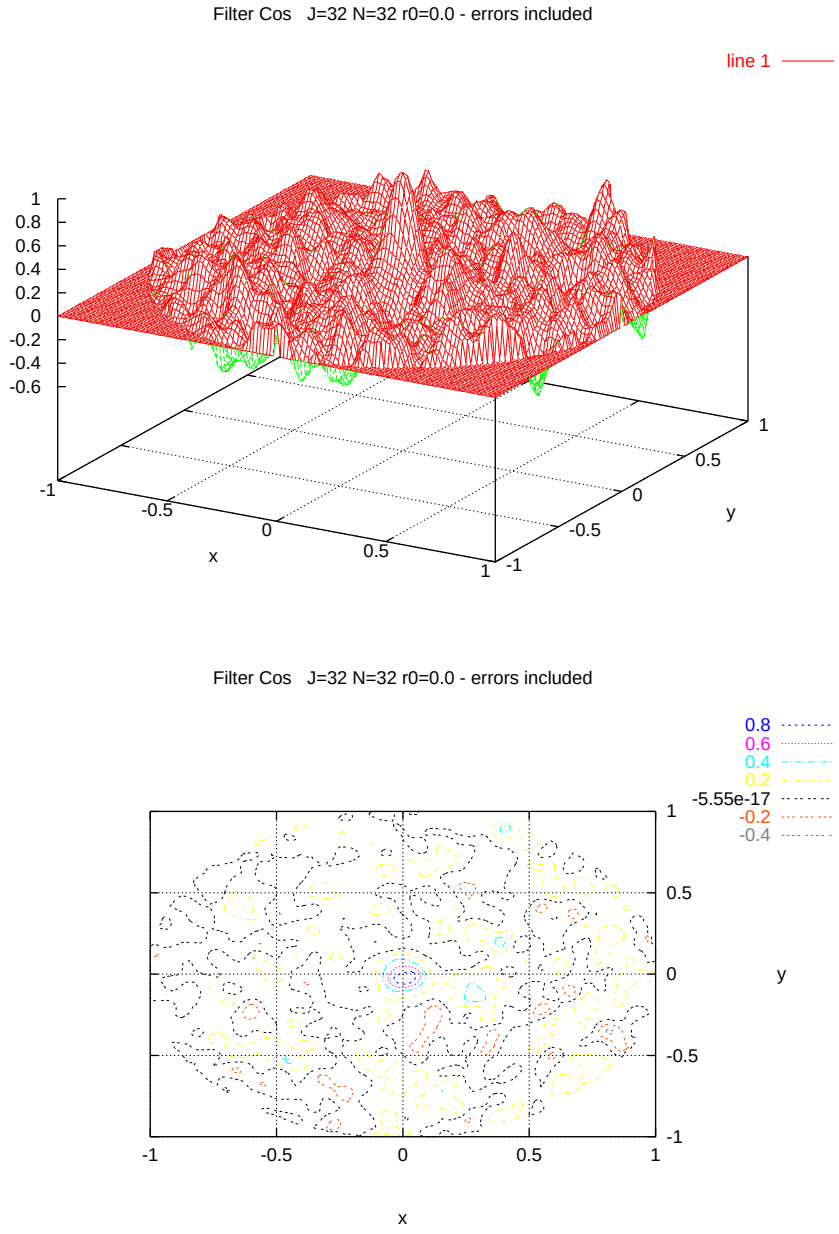


Figure B.28: Contour and mesh plots, $N = 32, J = 32, r_0 = 0$, error added.

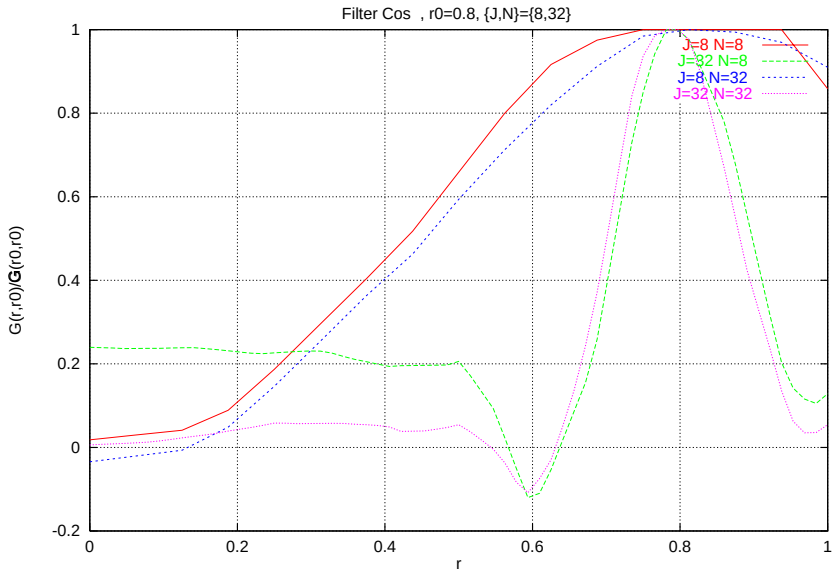


Figure B.29: $G(r, r_0)$ as fct. of r, point source positioned at $r_0 = 0.8$.

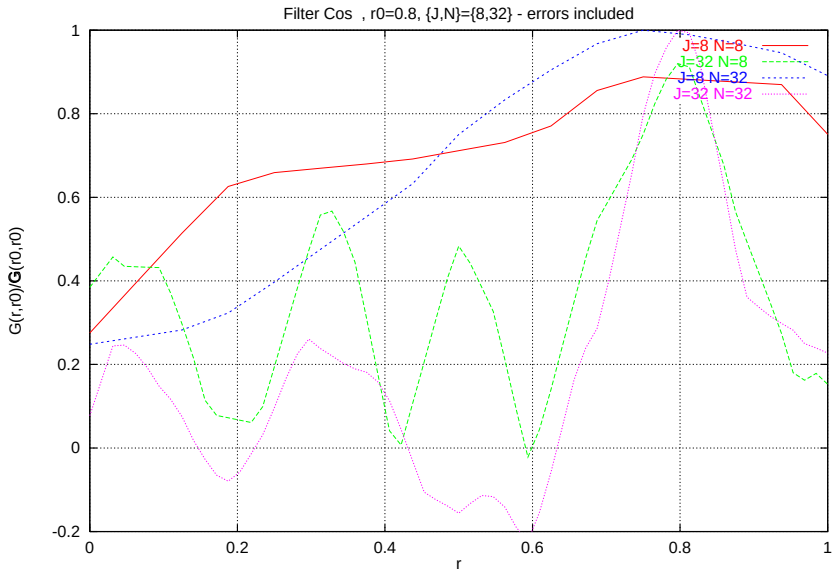


Figure B.30: As figure B.29, but errors added.

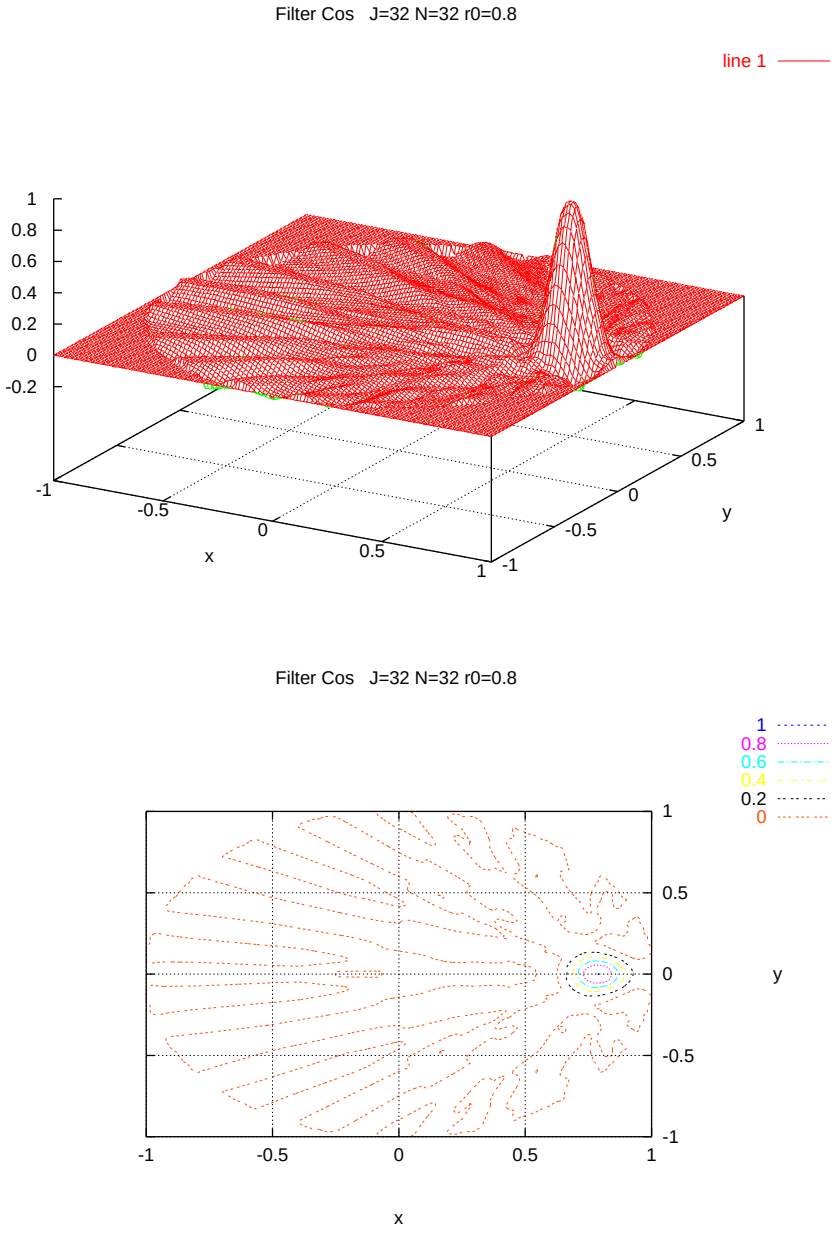


Figure B.31: Contour and mesh plots, $N = 32, J = 32, r_0 = 0.8$, no error.

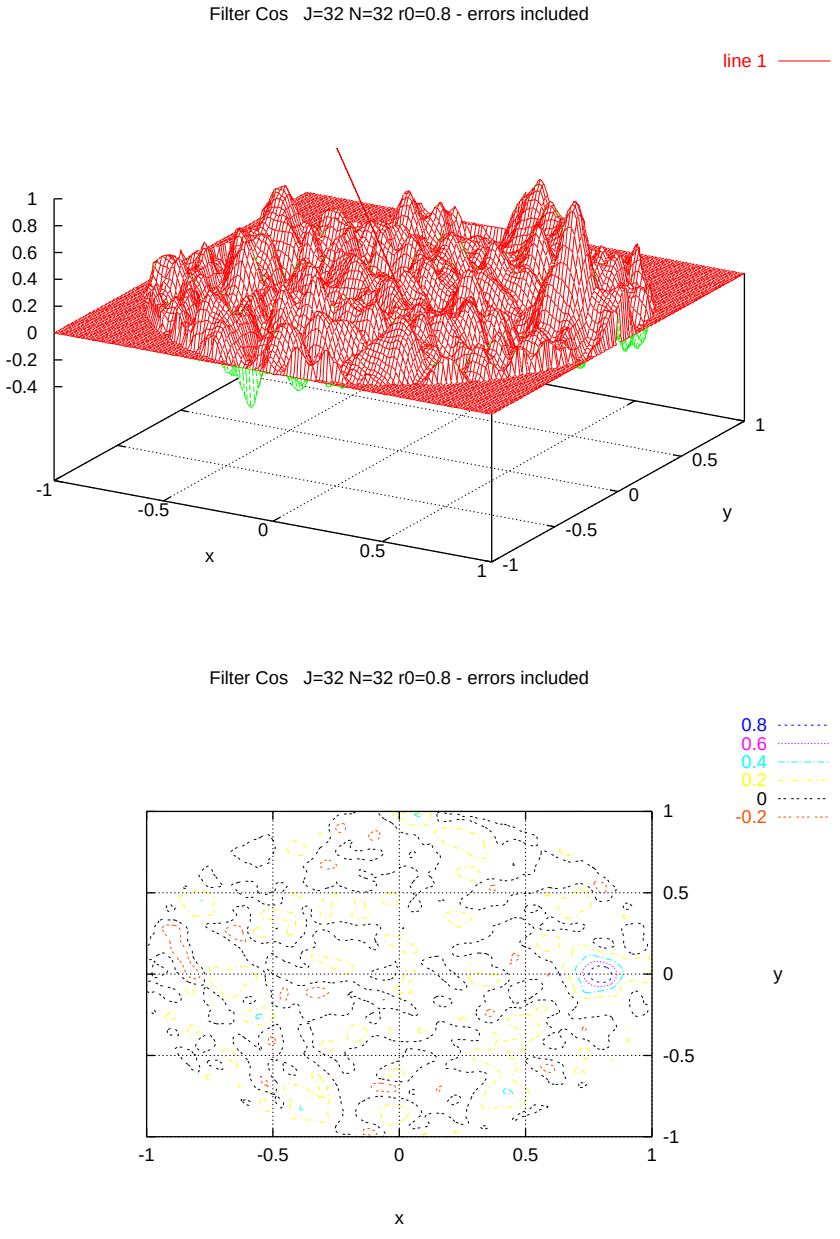


Figure B.32: Contour and mesh plots, $N = 32, J = 32, r_0 = 0.8$, error added.

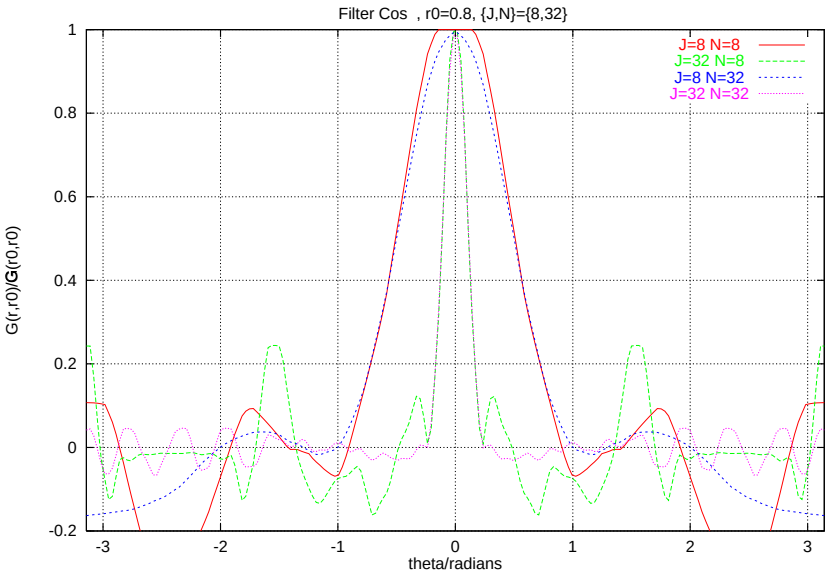


Figure B.33: $G(r, r_0)$ as fct. of θ, point source positioned at $r_0 = 0.8$.

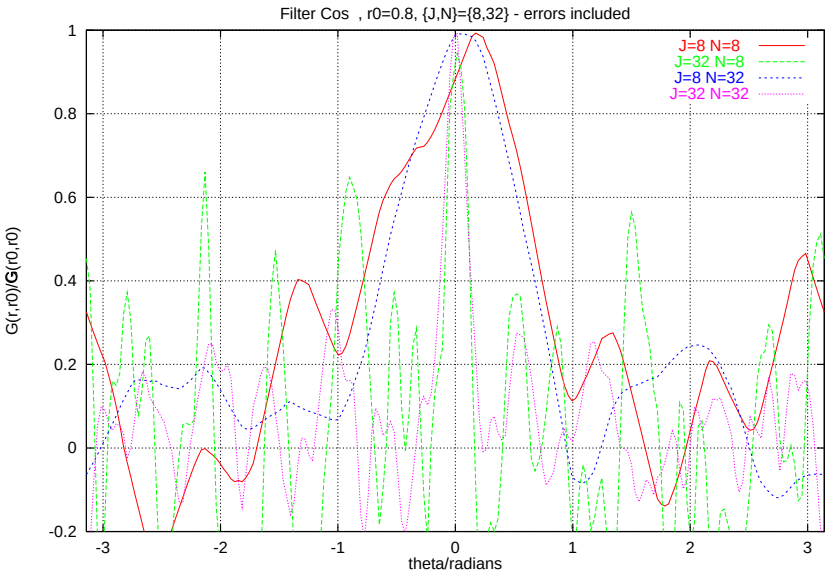


Figure B.34: As figure B.33, but errors added.

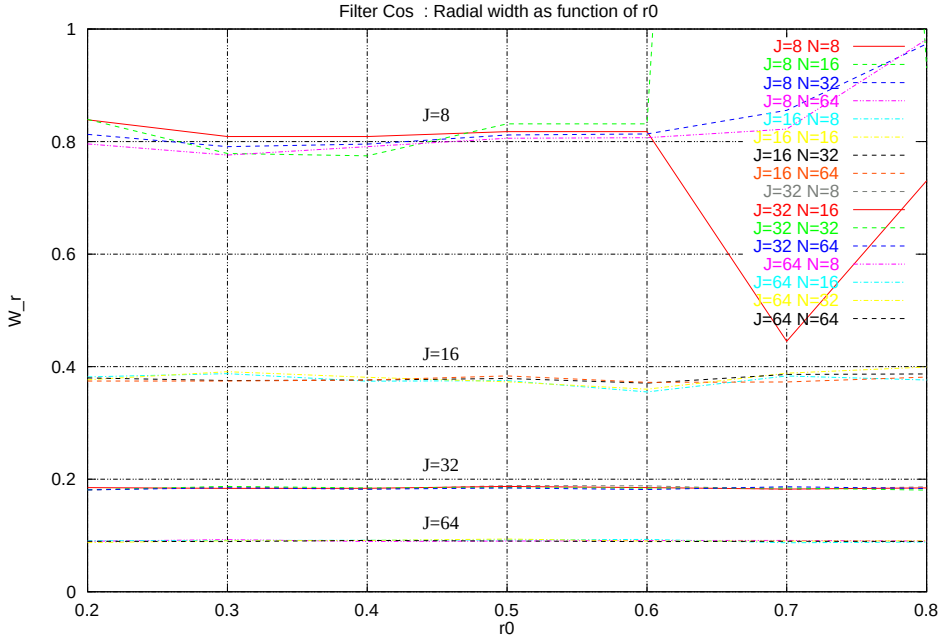


Figure B.35: Radial width W_t as fct. of r_0.

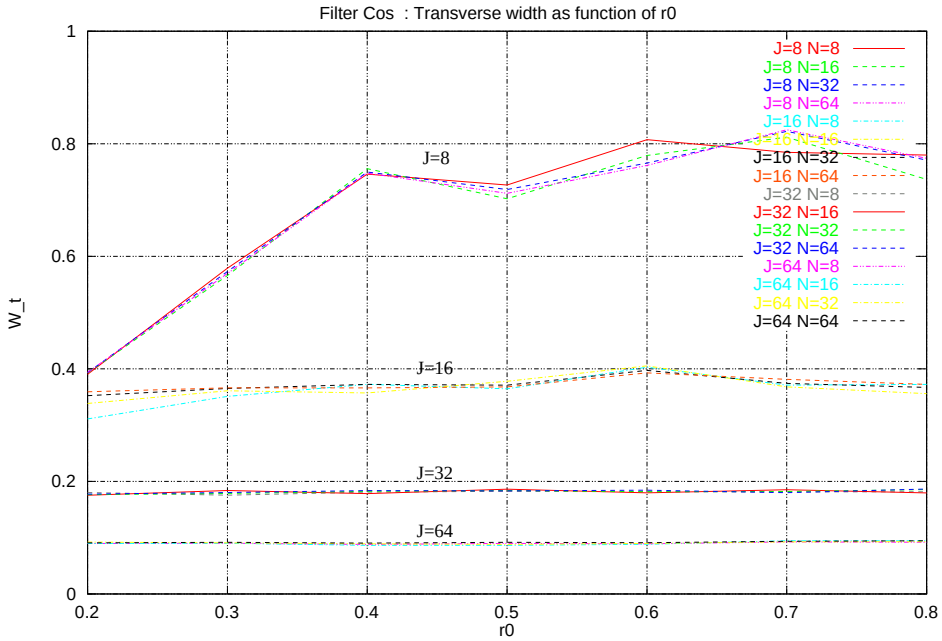


Figure B.36: Transverse width W_t as fct. of r_0.

Appendix C

Use of the cdrom

This appendix holds information on the contents and use of the enclosed cdrom.
On the cdrom you will find

- A more complete dataset than the figures presented within the thesis. The data are avaliable as Postscript figures in the `/data` directory. Read the file `README` in that directory for filename conventions.
- Source code for the used reconstruction program in `/src/c-code` and selected `Octave` scripts in the `/src/octave` directory.
- A hierachy of `HTML` pages visualising selected data, placed in `/visuals` on the cdrom. Among the presented data are animations of the dependence on the point spread function of selected parameters. The page layout is simple, and viewable using for instance Internet Explorer or Netscape Navigator.

A `HTML` page called `index.html` in the root catalog of the cdrom contains links to all of the above elements.

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